

Course
Electric Circuits

Module
Network Topology

UNIT-V

Network Topology

Definitions of Graph and Tree

Network Incidence matrices

Basic cut-set and tie-set matrices for planar networks

Loop and nodal methods of analysis of networks with dependent and independent voltage and current sources.

Duality and Dual networks

Network topology is a graphical representation of electric circuits.

It is useful for analyzing complex electric circuits by converting them into network graphs.

Network topology is also called as **Graph theory**.

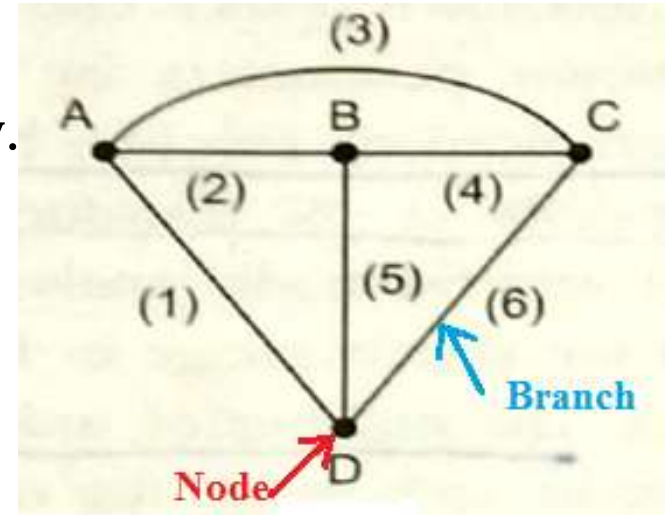
Graph

Network graph is simply called as **graph**.

It consists of a set of nodes connected by branches.

In a graph, node is a common point of two or more branches. Sometimes, only a single branch may connect to the node.

A branch is a line segment that connects two nodes.



Branch:

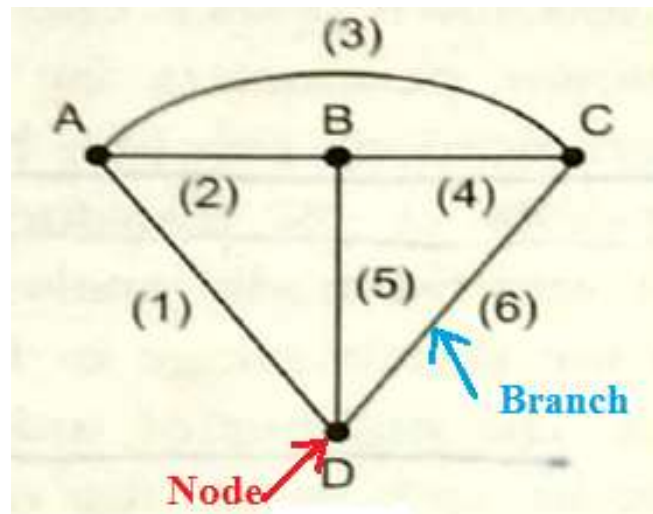
A branch is a line segment representing one network element or combination of element connected between two points.

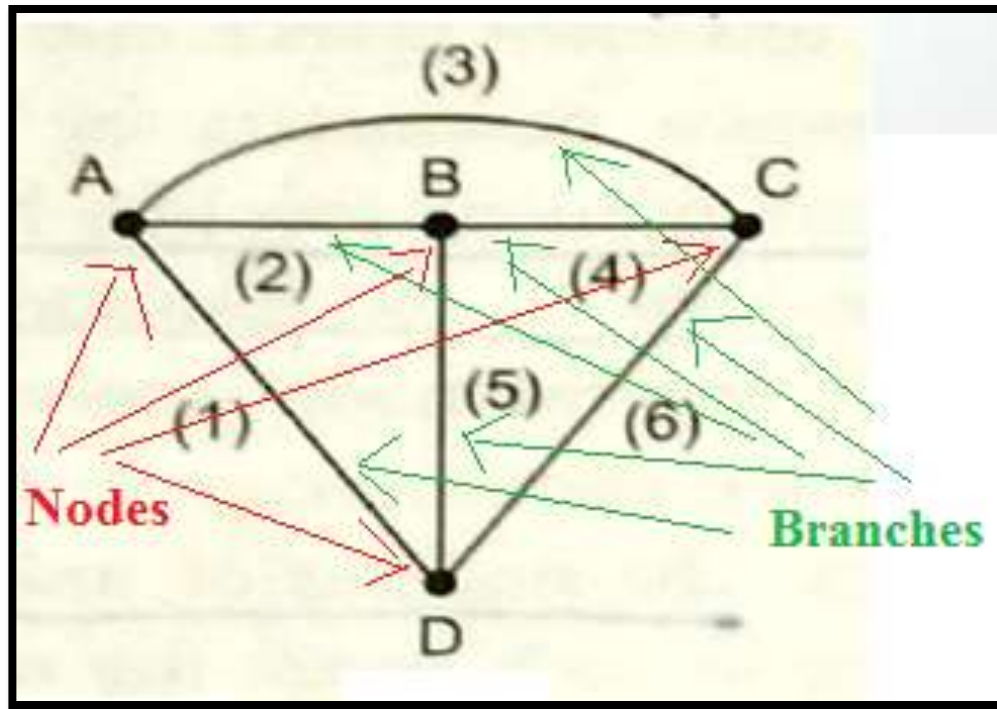
Node:

Node is an end point of a line segment and exists at the junction between two branch.

Degree of Node:

It is number of branches incident to it.





No. of Nodes in the graph : 4 $\rightarrow$ A, B, C and D

No. of Branches in the graph: 6 $\rightarrow$ (1), (2), (3), (4), (5) and (6).

Degree of Node-A is 3 (there are 3 branches incident to Node-A)

Similarly here all the remaining nodes have degree as 3.

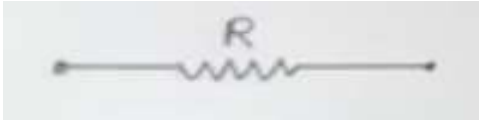
Any electric circuit or network can be converted into its equivalent **graph** by replacing the passive elements(R , L and C) and voltage sources(V) with short circuits and the current sources (I) with open circuits.

So the line segments in the graph represent the branches corresponding to either passive elements or voltage sources of electric circuit.

Representation of network elements to draw the graph:

SI NO	ELEMENT	REPRESENTATION
1	Resistor, R	short circuit (line segment)
2	Inductor, L	short circuit (line segment)
3	Capacitor, C	short circuit (line segment)
4	Voltage Source	short circuit (line segment)
5	Current Source	open circuit

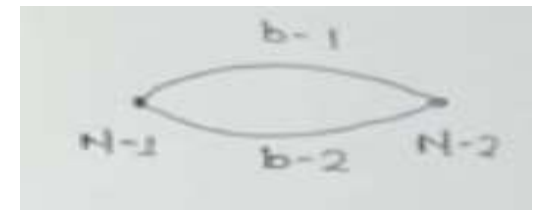
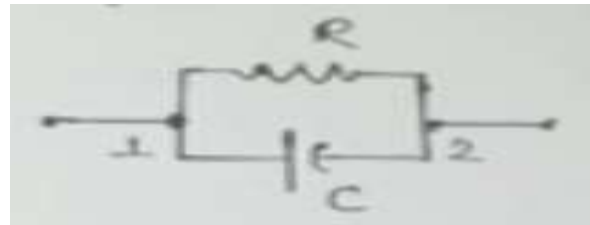
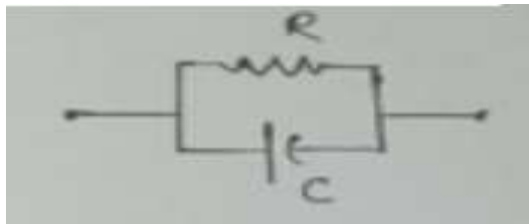
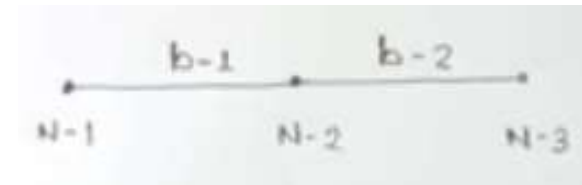
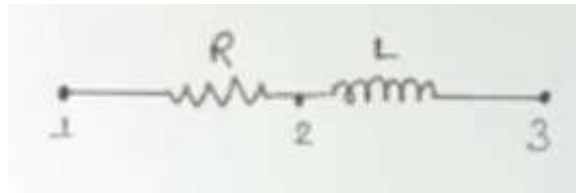
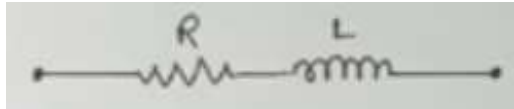
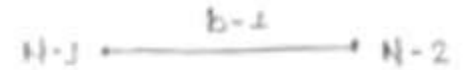
Element



Nodes Representation



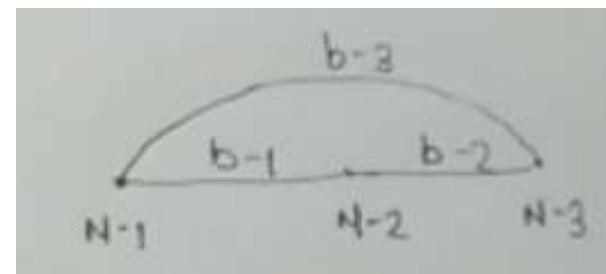
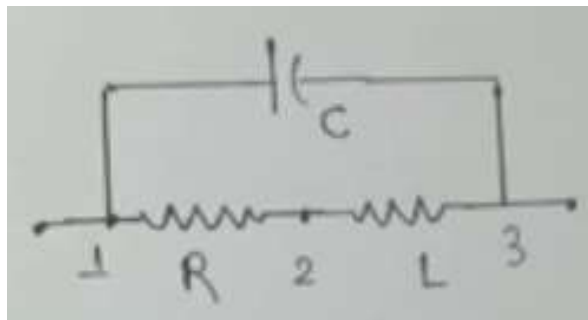
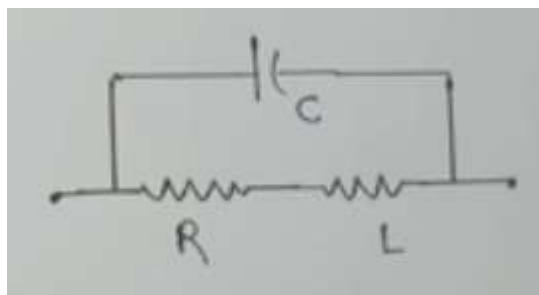
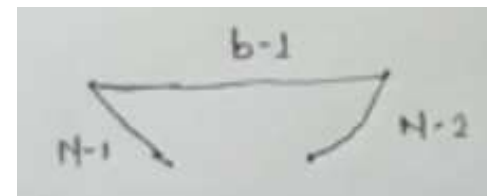
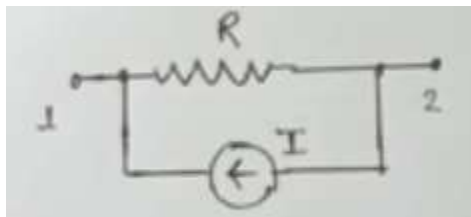
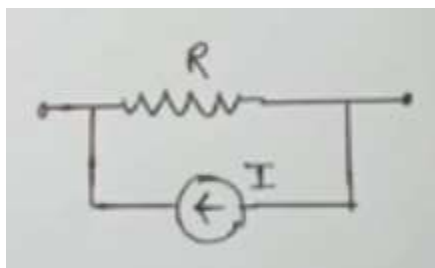
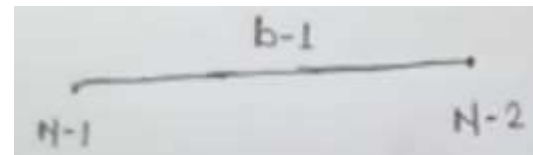
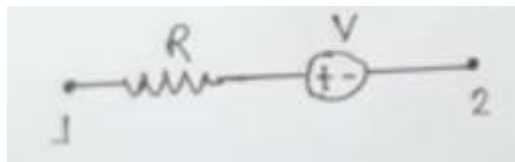
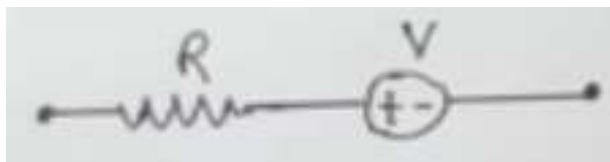
Graph



Element

Nodes Representation

Graph



Draw the graph for the following network shown in figure

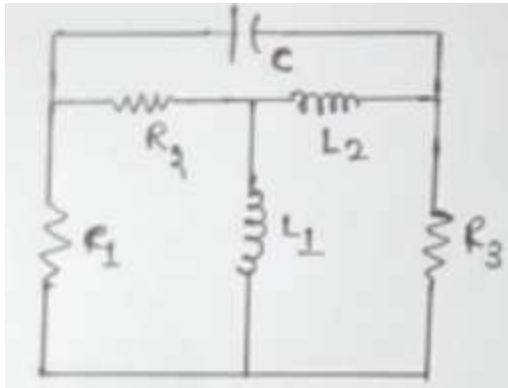
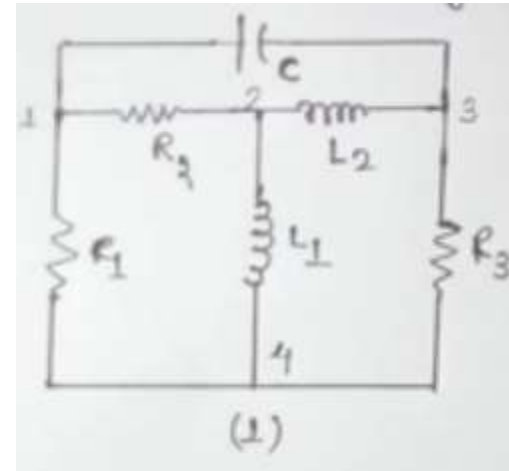
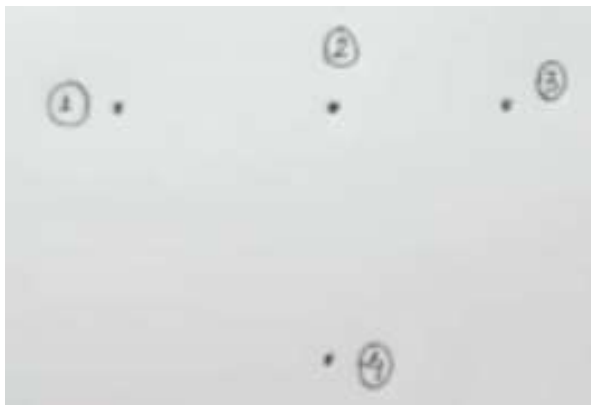


Fig: Network

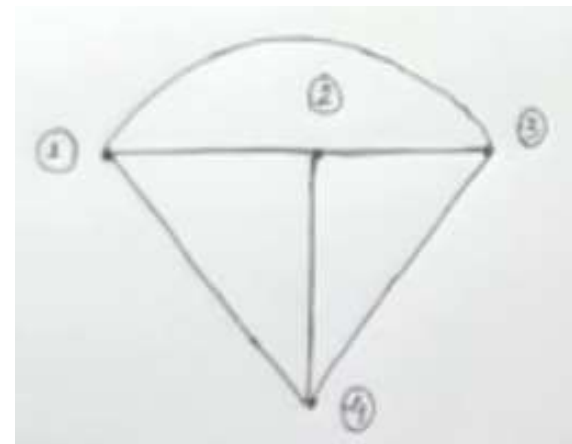
Step-1: Represent nodes in the network



Step-2: Takeout nodes to draw graph



Step-3: connect nodes as per network to complete graph



Draw the graph for the following network shown in figure

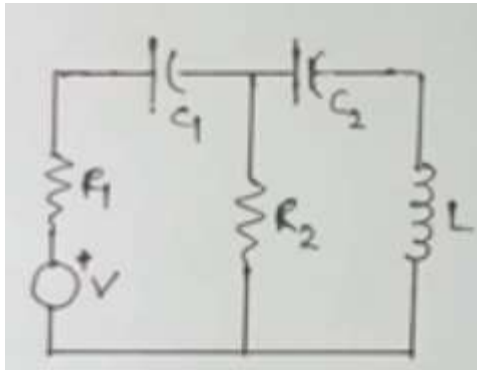
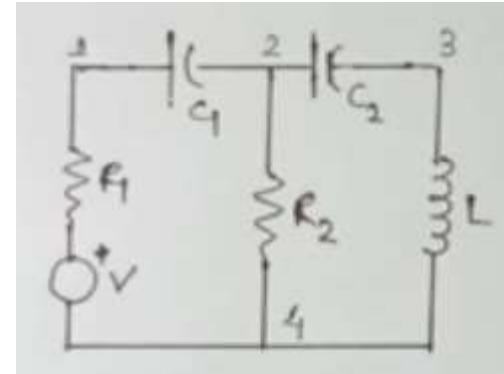
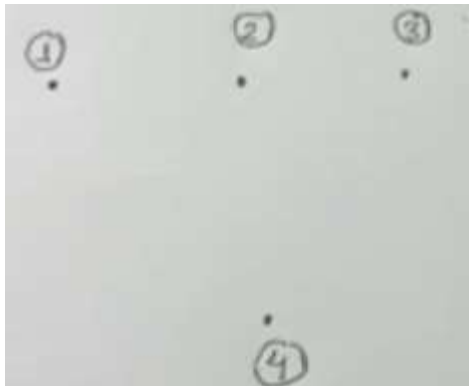


Fig: Network

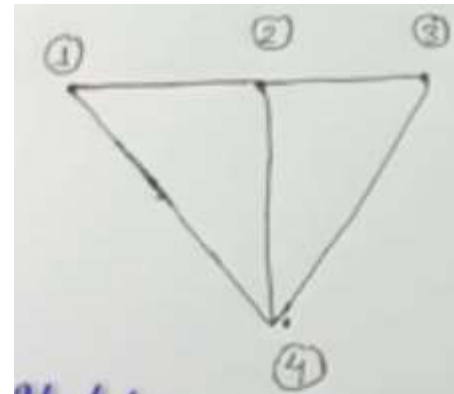
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Draw the graph for the following network shown in figure

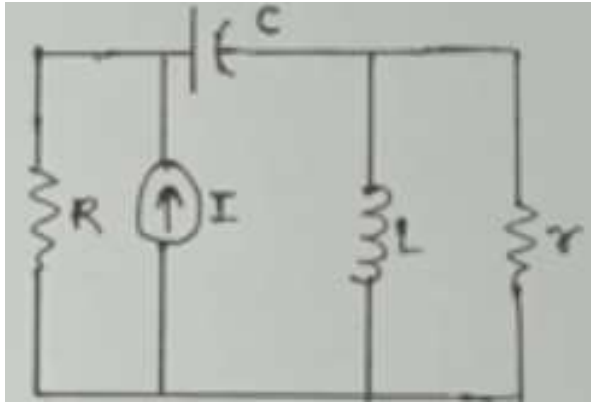
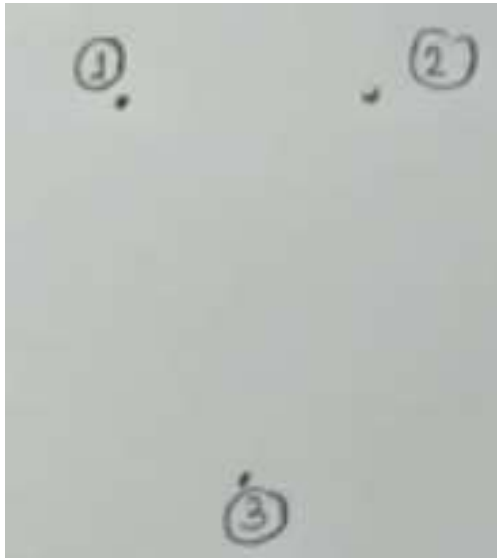
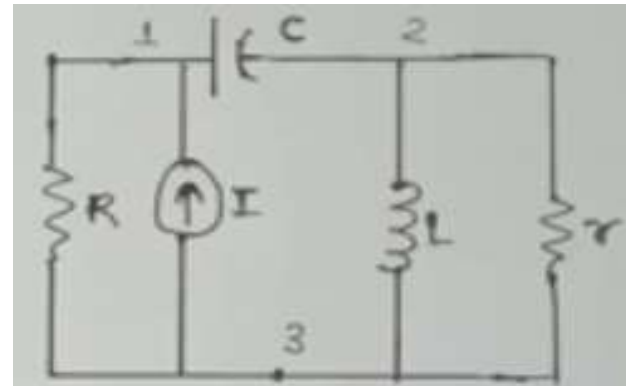


Fig: Network

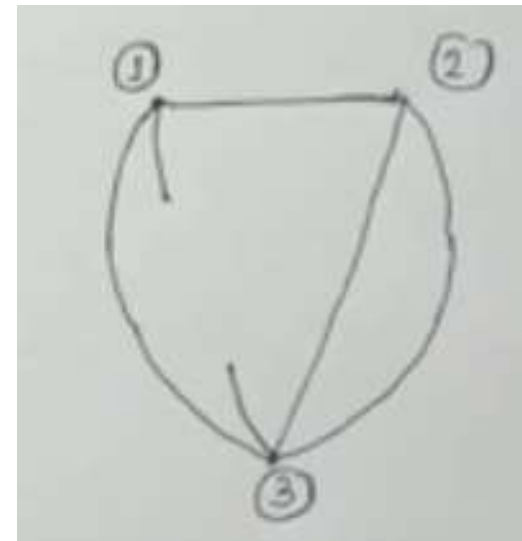
Step-2: Takeout nodes to draw graph

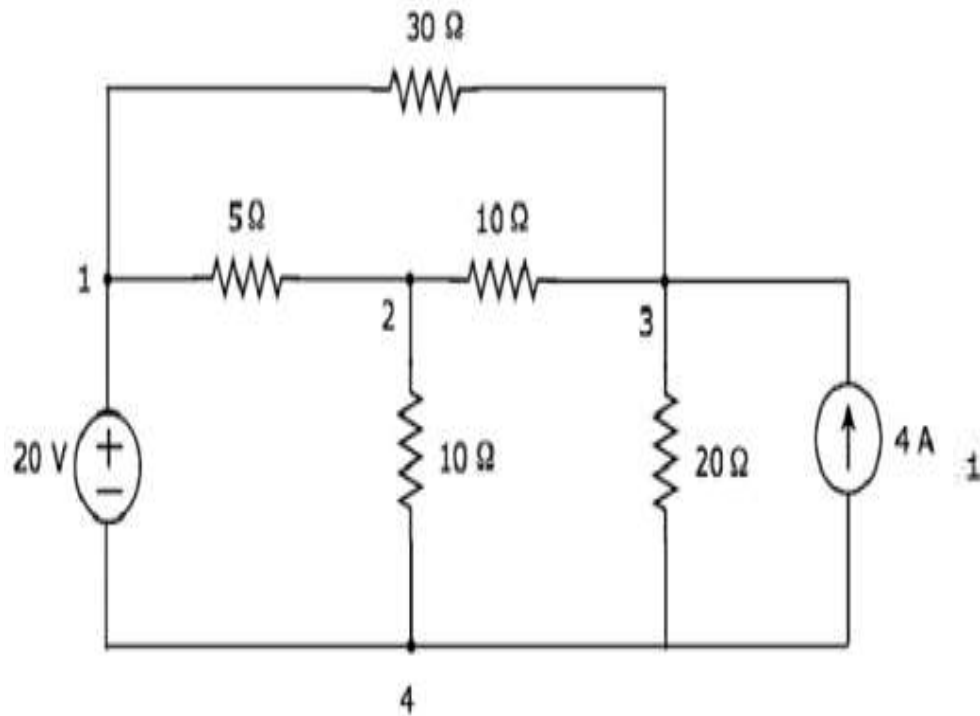


Step-1: Represent nodes in the network



Step-3: connect nodes as per network to complete graph

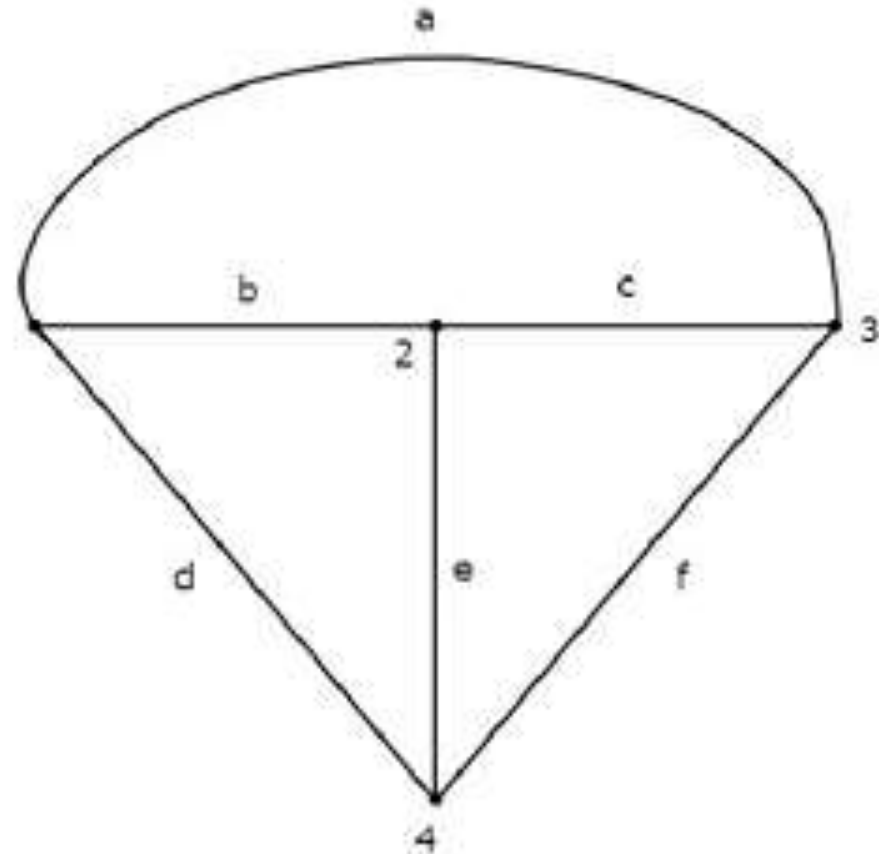




(a) Electrical Network

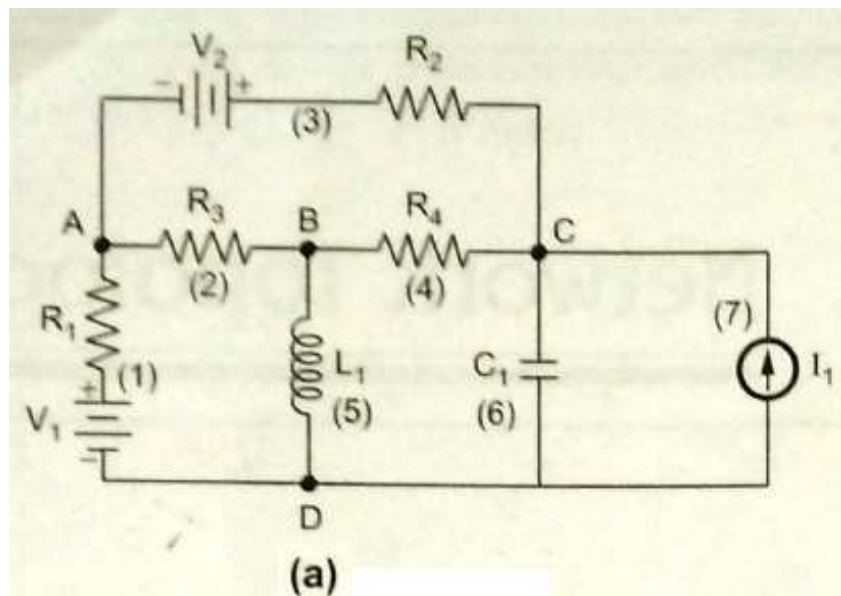
In the network: Nodes are 4 and Branches are 7 (voltage source, current source and 5 resistances)

In the graph : 4 nodes and 6 branches; since current source is made open.



(b) Graph of network

For the following electrical network shown in the figure (a) obtain the Graph?



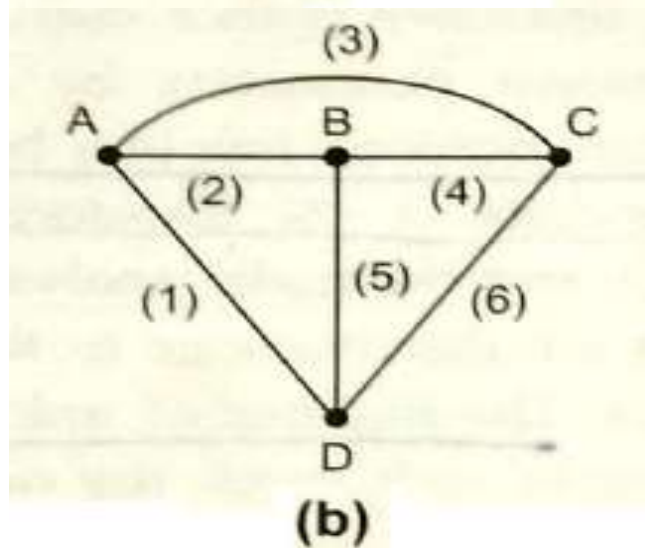
Voltage sources and passive elements are represented by lines (short circuit)

Current sources open circuit.

The Graph for the given network is shown in fig (b)

Conclusions:

1. The **number of nodes** present in a graph will be equal to the number of principal nodes present in an electric circuit.
2. The **number of branches** present in a graph will be less than or equal to the number of branches present in an electric circuit.



Types of Graphs

Following are the types of graphs

1. Connected and Unconnected Graphs
2. Oriented and Un-Oriented Graphs
3. Planar and Non-planar Graphs

1. Connected Graph

If there exists at least one branch between any of the two nodes of a graph, then it is called as a **connected graph**.

Each node in the connected graph will be having one or more branches that are connected to it. So, no node will present as isolated or separated.

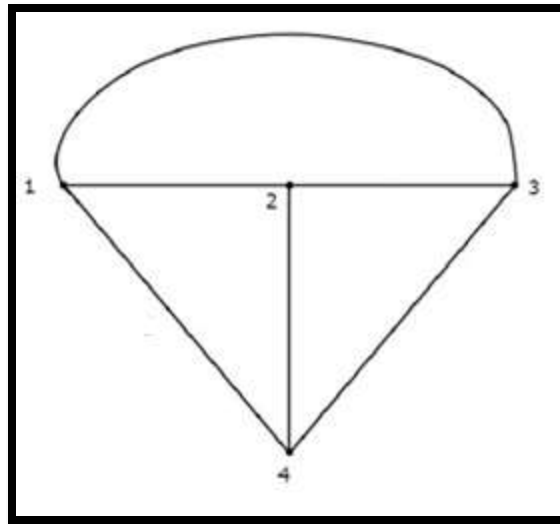


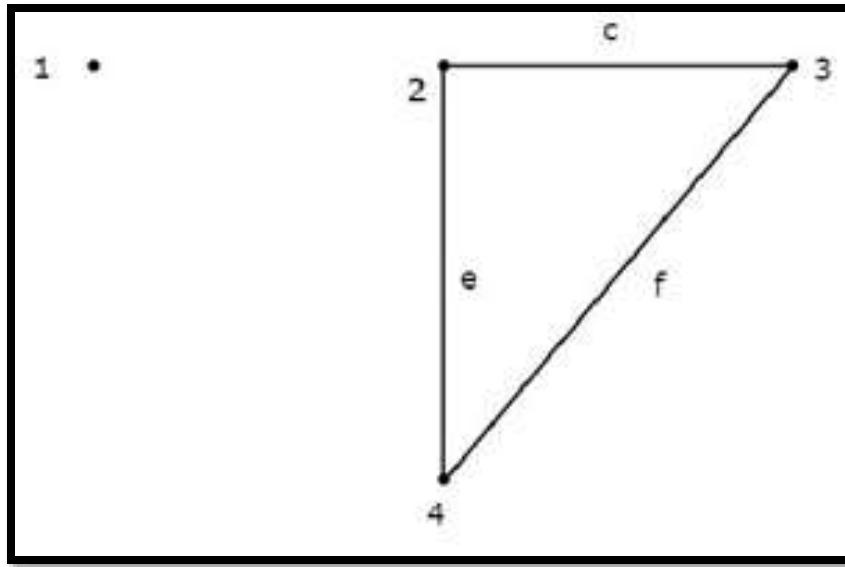
Fig: Connected Graph

2. Unconnected Graph

If there exists at least one node in the graph that remains unconnected by even single branch, then it is called as an **unconnected graph**.

So, there will be one or more isolated nodes in an unconnected graph.

Consider the graph shown in the following figure.



In this graph, the nodes 2, 3, and 4 are connected by two branches each. But, not even a single branch has been connected to the **node 1**.

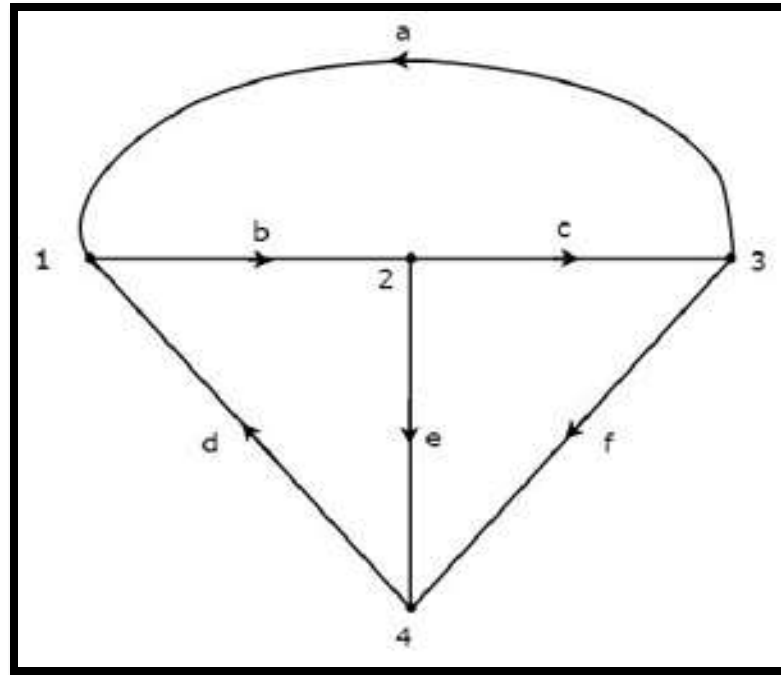
So, the node 1 becomes an **isolated node**. Hence, the above graph is an **unconnected graph**.

3. Oriented Graph

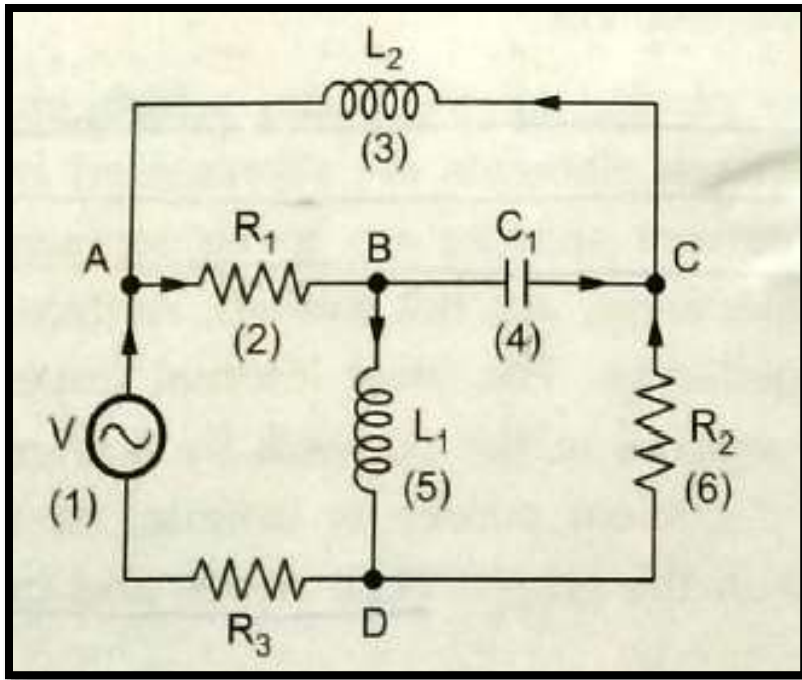
If all the branches of a graph are represented with arrows, then that graph is called as a **oriented graph**.

These arrows indicate the direction of current flow in each branch.

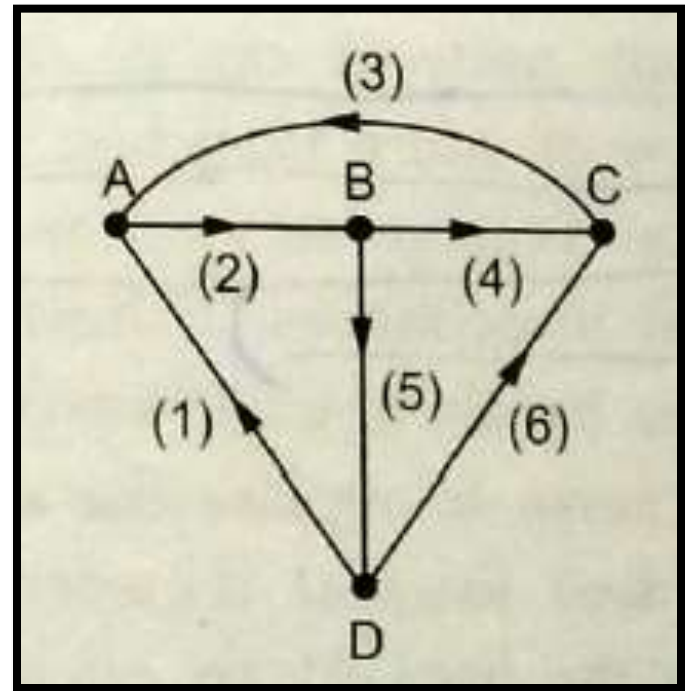
Consider the graph shown in the following figure.



This graph is **oriented graph**, the direction of current flow is represented with an arrow in each branch.



(a) Network



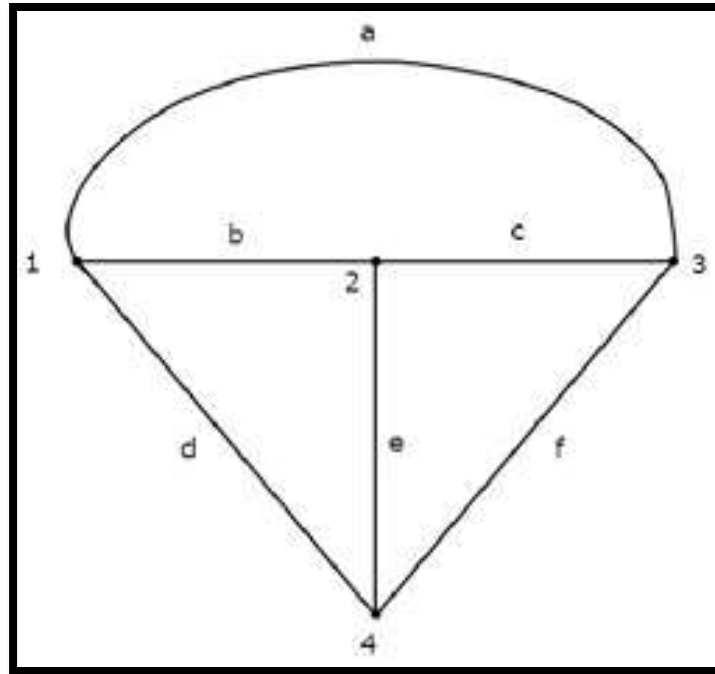
(b) Oriented Graph

Fig: Network and Oriented Graph

4. Un-oriented Graph

If the branches of a graph are not represented with arrows, then that graph is called as an **un-oriented graph**.

The graph that was shown in the following example is an **un-oriented graph**, because there are no arrows on the branches of that graph.



5. Planar Graph:

A Planar Graph is a graph drawn on a two-dimensional plane so that no two branches intersect at a point which is not a node.

The graphs shown in the figure are planar graphs.

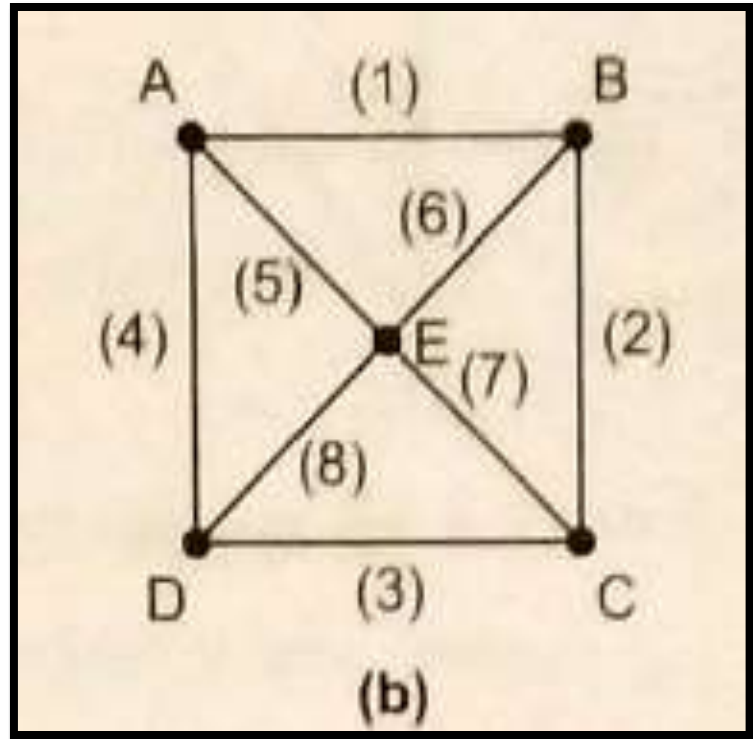
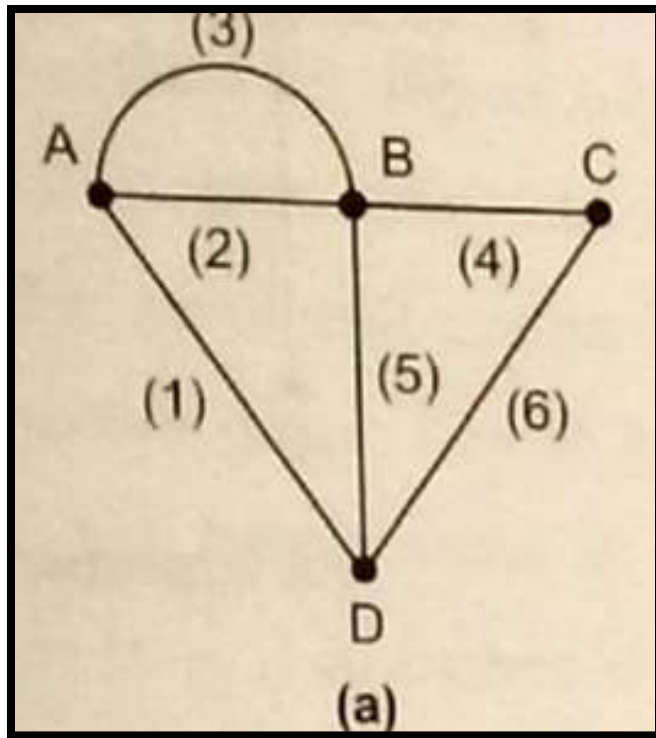


Fig: Planar Graphs

6. Non- Planar Graph:

It is a graph drawn on a two dimensional plane such that two or more branches intersect at a point other than node on a graph.

The graphs shown in the figure are planar graphs.

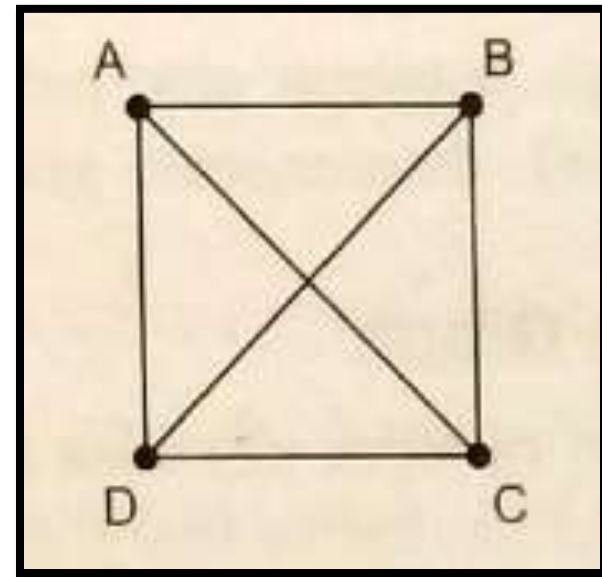
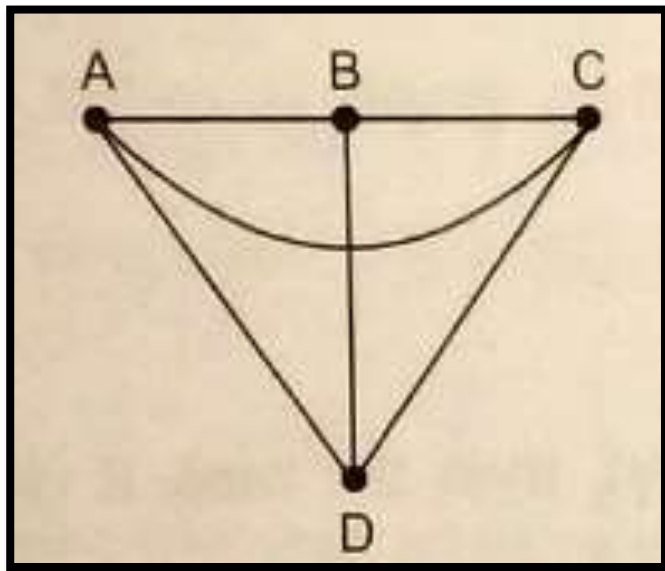


Fig: Non-Planar Graphs

Subgraph

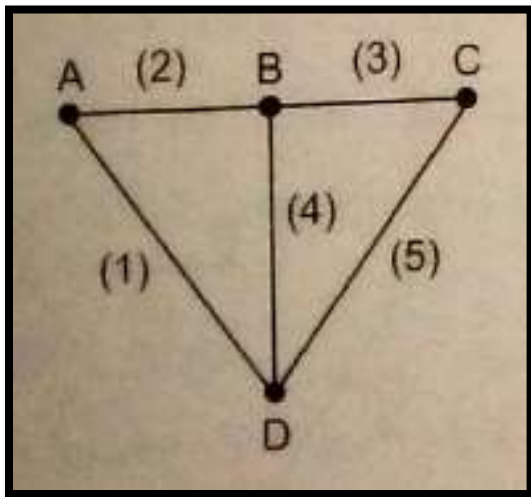
A subgraph is a subset of branches and nodes of a graph.

A subgraph can be of two types (1) proper subgraph and (2) improper subgraph.

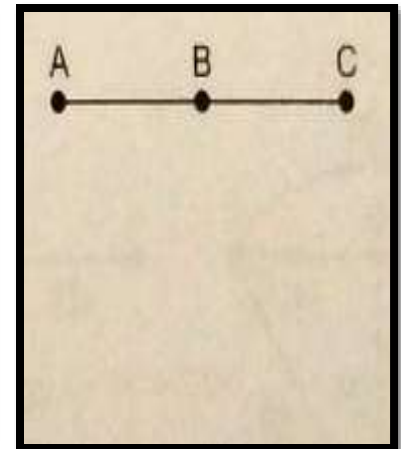
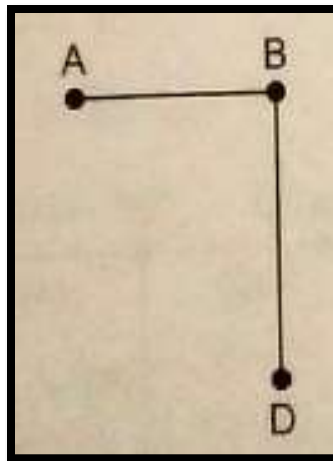
proper subgraph:

If the subgraph contains **branches and nodes less than** those on the graph, then such a subgraph is called proper subgraph.

The graph and its proper subgraphs are shown in figure.



(a) Graph

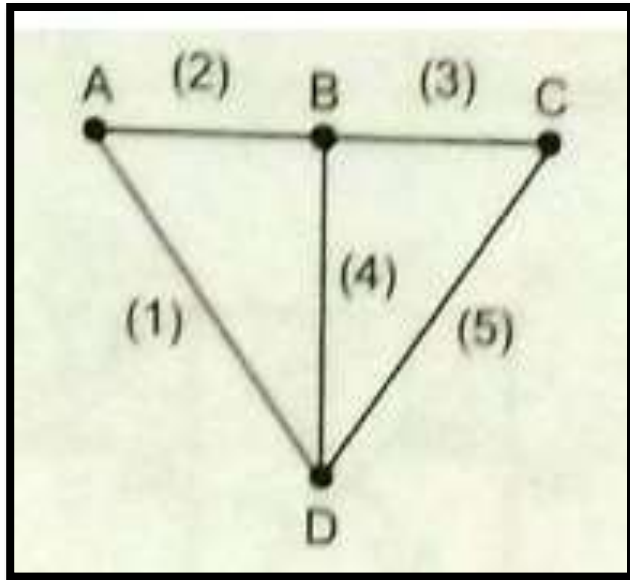


(b) Subgraphs

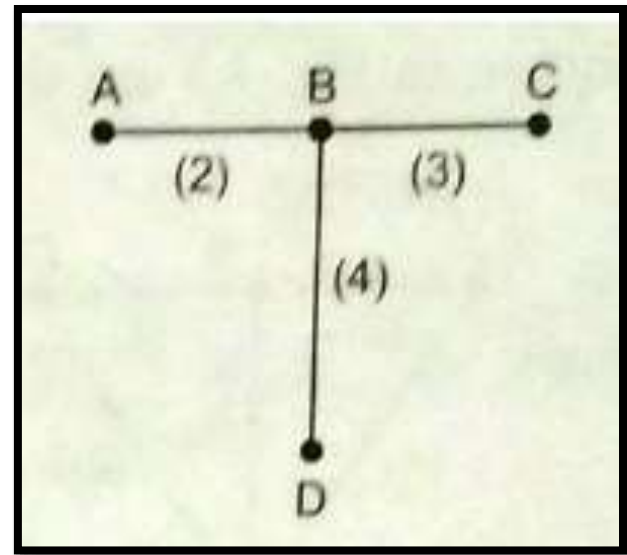
Improper subgraph:

If a subgraph contains **all the nodes and less branches** of a graph then a subgraph is called improper subgraph.

A graph and its improper subgraphs are as shown in figure.



(a) Graph



(b) Improper Graph

Fig: Graph and improper graph

Tree

Tree is a connected subgraph of a given graph, which contains **all the nodes** of a graph. But, there **should not be any loop** in that subgraph.

The branches of a tree are called as **twigs** and represented with solid lines.

If there 'n' number of nodes in the graph then the **number of branches** in a Tree equal to $n - 1$.

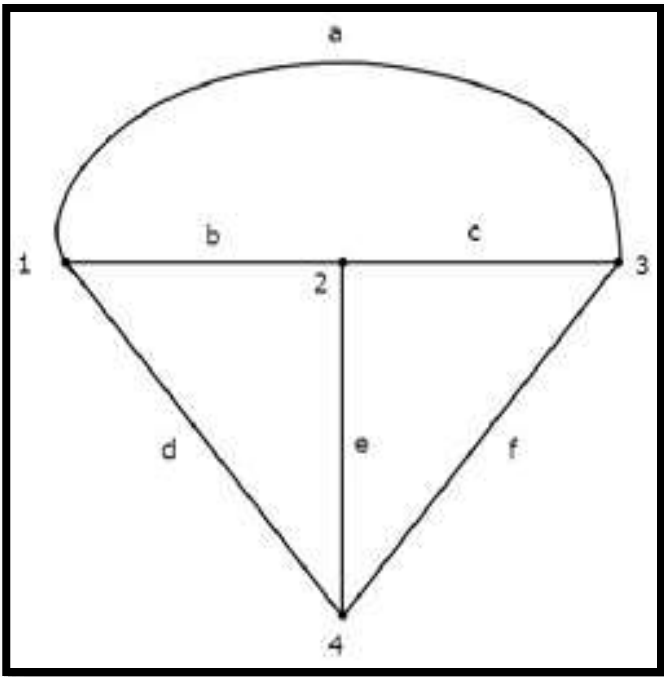


Fig (a) : Graph

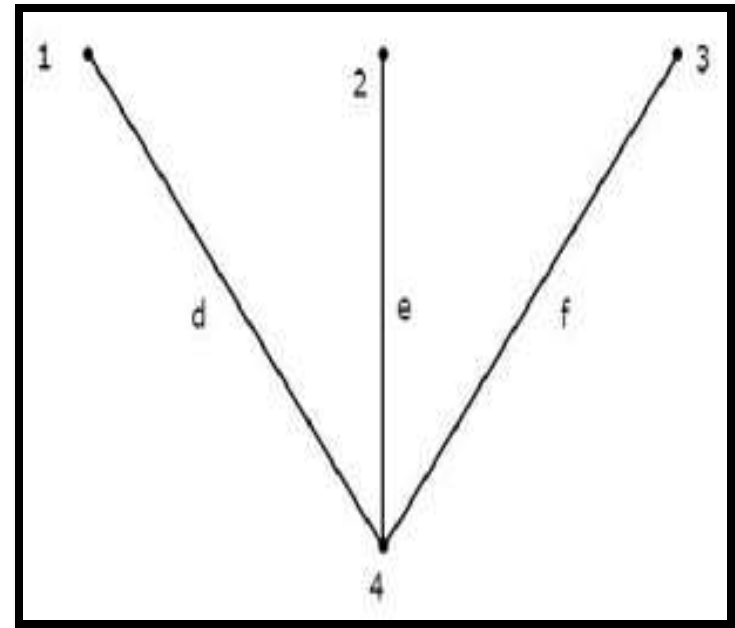


Fig (b): one of Tree of Graph

The connected subgraph in fig(b) contains all the four nodes of the given graph but there is no loop. Hence, it is a **Tree**.

Draw the different possible trees for the following graph in the figure.

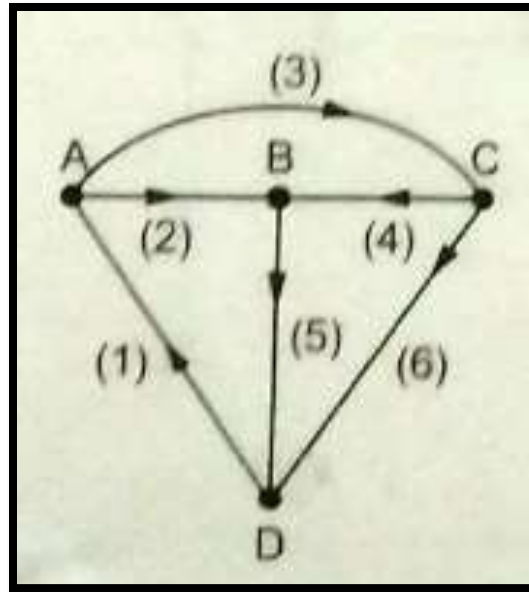


Fig: Graph

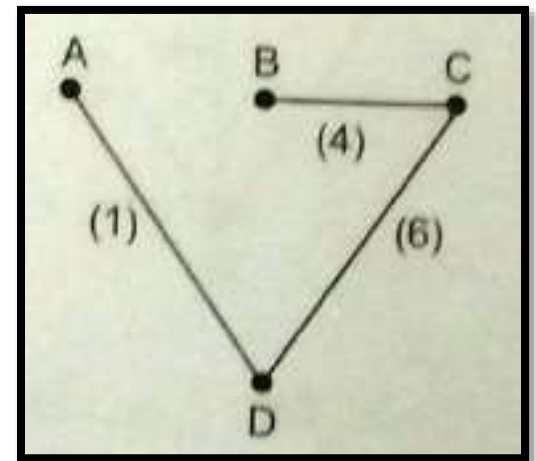
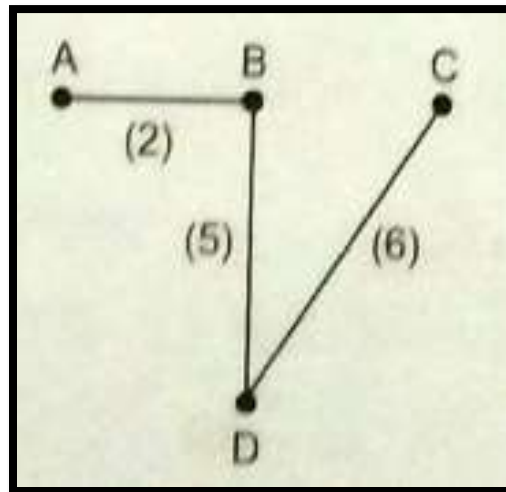
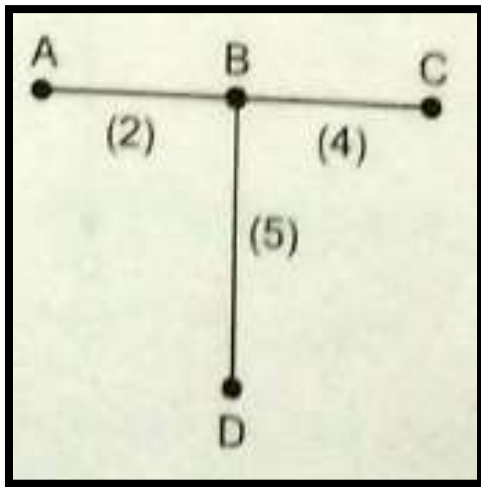


Fig: possible trees of given Graph

Draw the different possible trees for the following graph in the figure.

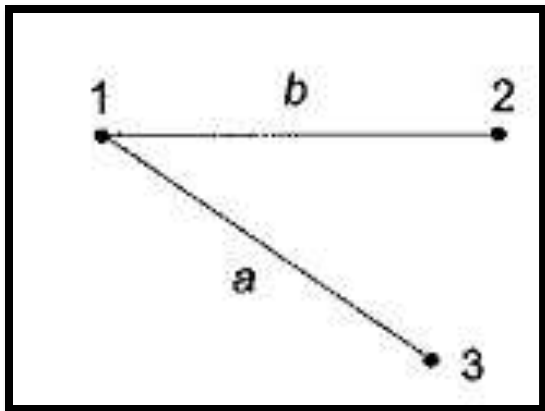
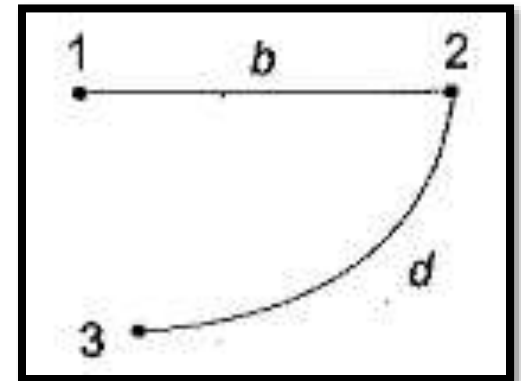
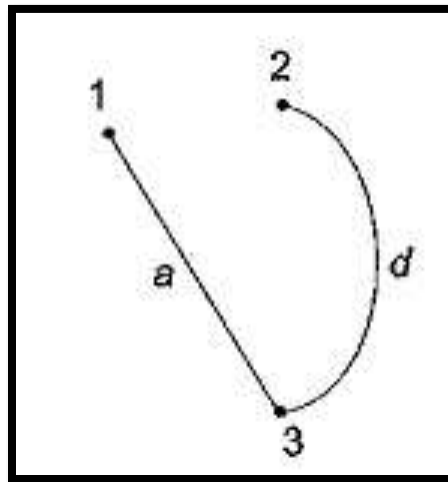
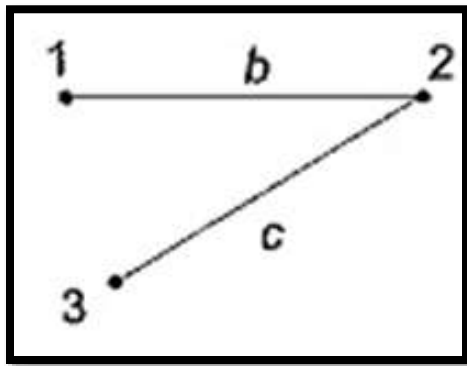
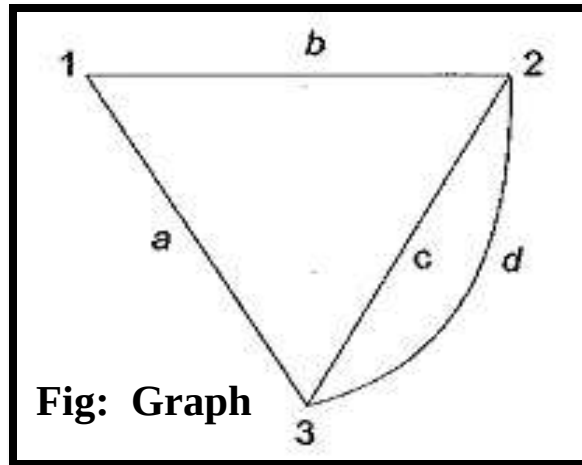
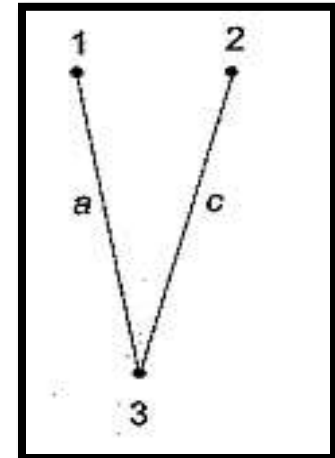


Fig: Possible trees of Graph



Co-Tree

Co-Tree is a subgraph, which is formed with the branches that are removed while forming a Tree.

It is called as **Complement** of a Tree.

The co-tree branches are called as **links or chords**.

In general, the links are represented with dotted lines.

The **Co-Tree** corresponding to the above Tree is shown in the following figure.

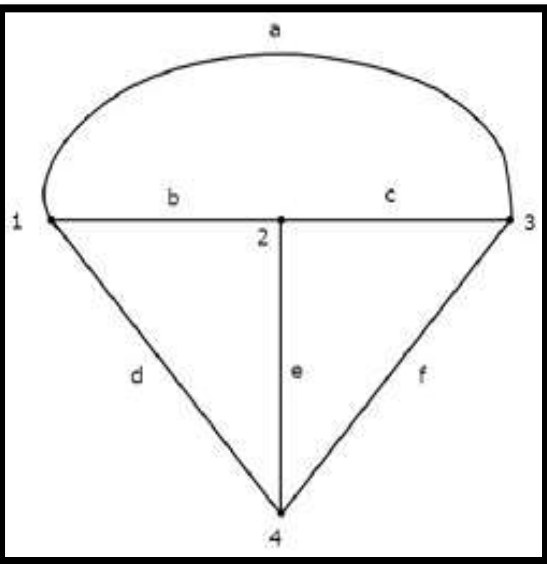


Fig (a) : Graph

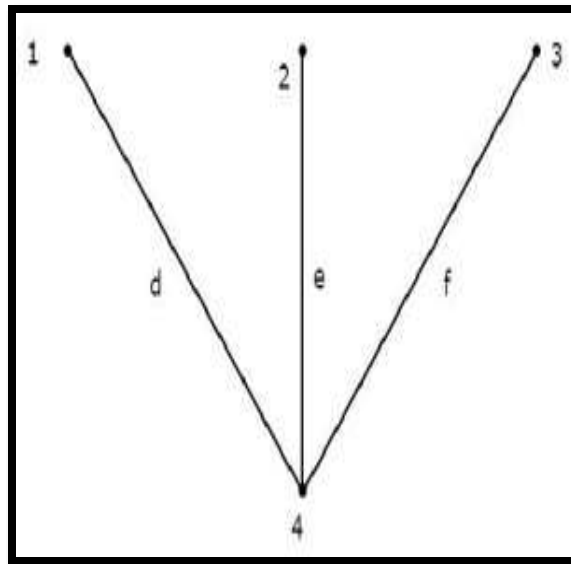
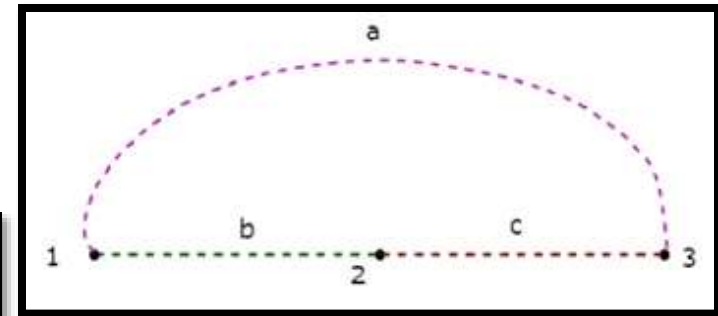


Fig (b): one of Tree of Graph



**Fig (c) : Co-tree
corresponding to tree in fig (b)**

If we combine a Tree and its corresponding Co-Tree, then we will get the **original graph** as shown below.

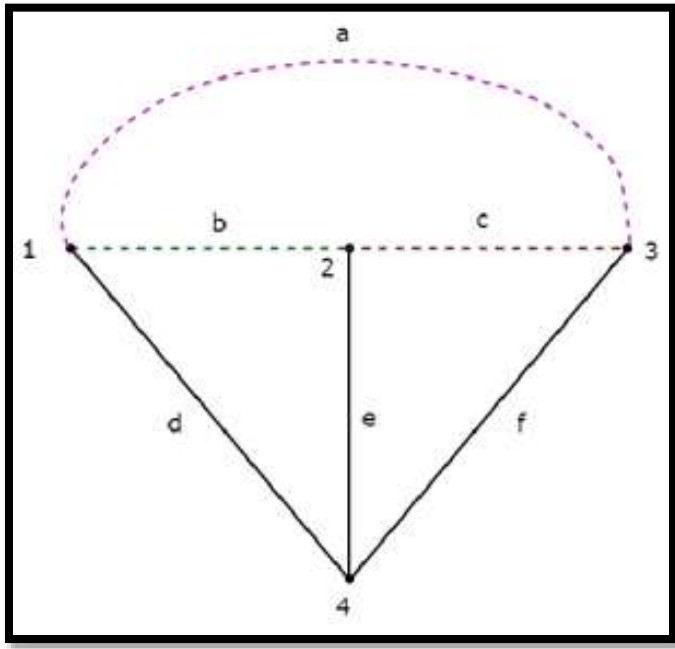


Fig: Graph with a tree and corresponding co-tree

The Tree branches d, e & f are represented with solid lines. The Co-Tree branches a, b & c are represented with dashed lines.

This Co-Tree has only three nodes instead of four nodes of the given graph, because Node 4 is isolated from the above Co-Tree. Therefore, the Co-Tree need not be a connected subgraph.

This Co-Tree has three branches and they form a loop. The number of branches that are present in a co-tree will be equal to the difference between the number of branches of a given graph and the number of twigs.

Mathematically, it can be written as

$$l = b - (n - 1)$$

Where,

l number of links

b number of branches in given graph

n number of nodes in given graph

Draw the possible tree and the corresponding co-tree for the given graph shown below

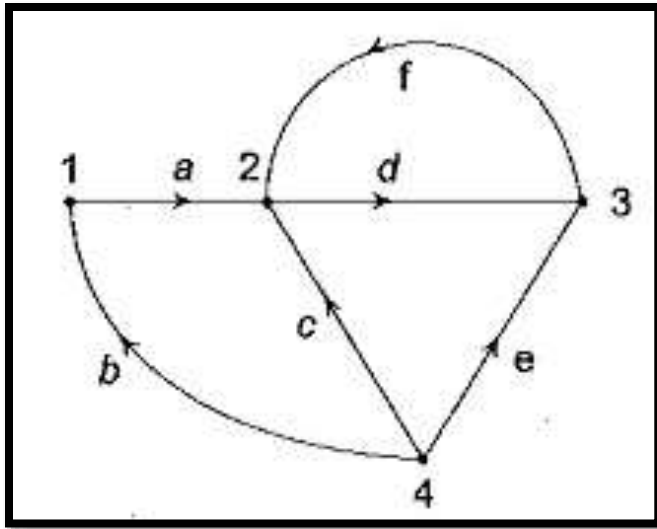


Fig: Graph

No. of Nodes , $n = 4$ (1, 2, 3 and 4)

No. of Branches, $b = 6$ (a, b, c, d, e and f)

No. of Twigs = $n-1 = 3$

Therefore NO. of links, $l = b-(n+1) = 3$

For the selected tree
The twigs are a, c
and e
and then the links for
the co-tree are b, d and
f.

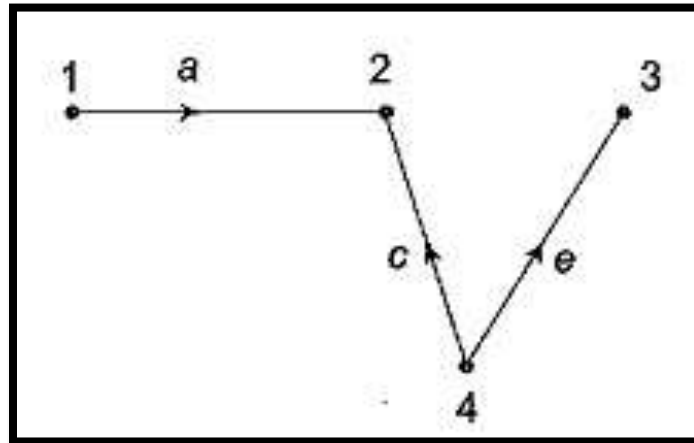


Fig: Selected Tree

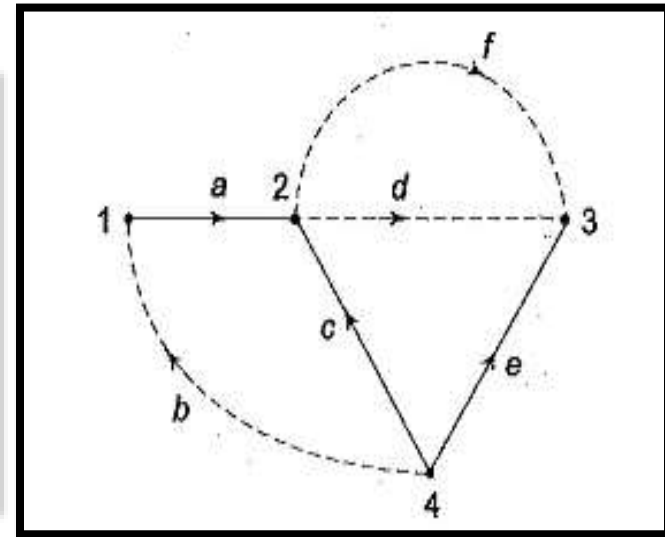


Fig: Co-Tree

Matrices Associated with Network Graphs

Following are the three matrices that are used in Graph theory.

1. Incidence Matrix
2. Fundamental Loop (Tie-set) Matrix
3. Fundamental Cut set Matrix

Incidence Matrix

A graph consists of a set of nodes and those are connected by some branches. So, the connecting of branches to a node is called as incidence. An Incidence Matrix represents the graph of a given electric circuit or network.

It is possible to draw the graph of that same electric circuit or network from the **incidence matrix**.

Incidence matrix is represented with the letter A. It is also called as node to branch incidence matrix or **node incidence matrix**.

If there are 'n' nodes and 'b' branches are present in a **directed graph**, then the incidence matrix will have 'n' rows and 'b' columns.

Rows and columns are corresponding to the nodes and branches of a directed graph. Hence, the **order** of incidence matrix will be $\mathbf{n} \times \mathbf{b}$.

Procedure to find Incidence Matrix

Follow these steps in order to find the incidence matrix of directed graph

1. Select a node at a time of the given directed graph and fill the values of the elements of incidence matrix corresponding to that node in a row.
2. The **elements of incidence matrix** will be having one of these three values, +1, -1 and 0.

If the branch current is leaving from a selected node, then the value of the element will be +1.

If the branch current is entering towards a selected node, then the value of the element will be -1.

If the branch current neither enters at a selected node nor leaves from a selected node, then the value of element will be 0.

Repeat the above step for all the nodes of the given directed graph.

In matrix A with n rows and b columns an entry a_{ij} in the i^{th} row and j^{th} column has the following values.

$$\left. \begin{aligned} a_{ij} &= 1, \text{ if the } j^{\text{th}} \text{ branch is incident to and oriented away from the } i^{\text{th}} \text{ node} \\ a_{ij} &= -1, \text{ if the } j^{\text{th}} \text{ branch is incident to and oriented towards the } i^{\text{th}} \text{ node.} \\ a_{ij} &= 0, \text{ if the } j^{\text{th}} \text{ branch is not incident to the } i^{\text{th}} \text{ node.} \end{aligned} \right\}$$

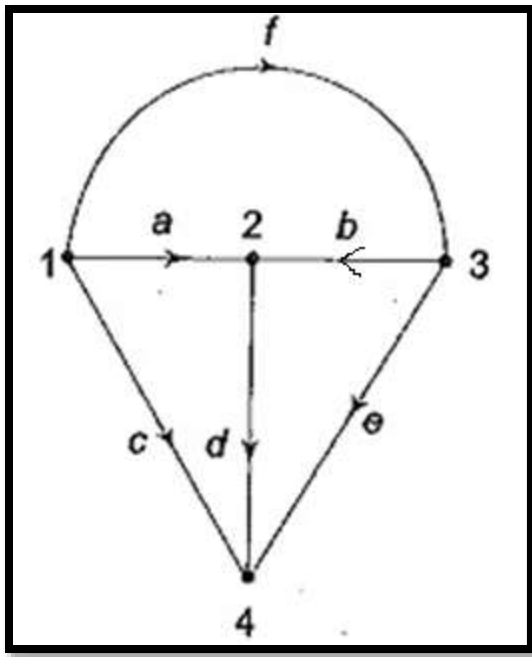


Fig: Graph of a network

	Nodes	Branches →					
	↓	a	b	c	d	e	f
$A =$	1	1	0	1	0	0	1
	2	-1	-1	0	-1	0	0
	3	0	1	0	0	1	-1
	4	0	0	-1	1	-1	0

Fig: Incidence matrix of graph

Consider the following directed graph. Obtain the incidence matrix?

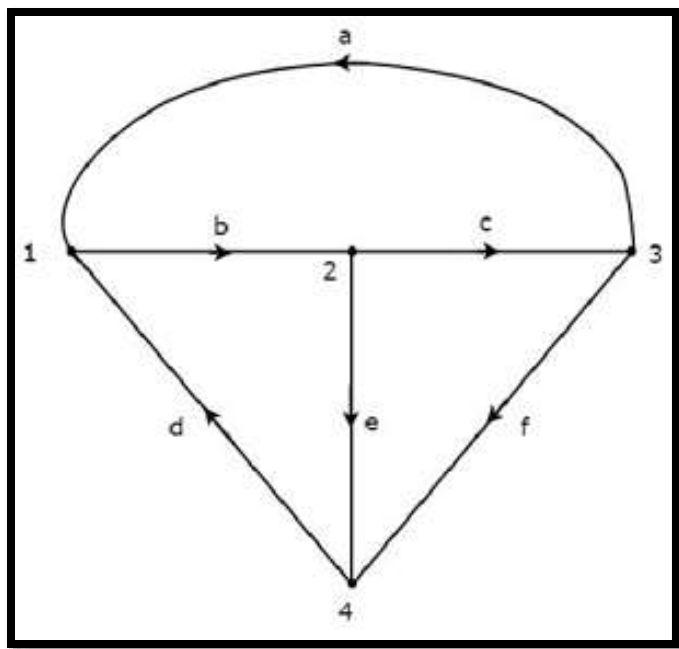


Fig: Graph

From the graph: No. of nodes = 4 and
No. of branches = 6

The rows and columns of the above matrix represents the nodes and branches of given directed graph. Therefore the order of incidence matrix is 4×6 .

If the branch current is leaving from a selected node, then the value of the element will be +1.

If the branch current is entering towards a selected node, then the value of the element will be -1.

If the branch current neither enters at a selected node nor leaves from a selected node, then the value of element will be 0.

The **incidence matrix** corresponding to the above directed graph will be

$$A = \begin{bmatrix} -1 & 1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 & -1 \end{bmatrix}$$

By observing the above incidence matrix, we can conclude that the **summation** of column elements of incidence matrix is equal to zero. That means, a branch current leaves from one node and enters at another single node only.

Note: If the given graph is an un-directed type, then convert it into a directed graph by representing the arrows on each branch of it.

We can consider the arbitrary direction of current flow in each branch.

For the following graph find the incidence matrix?

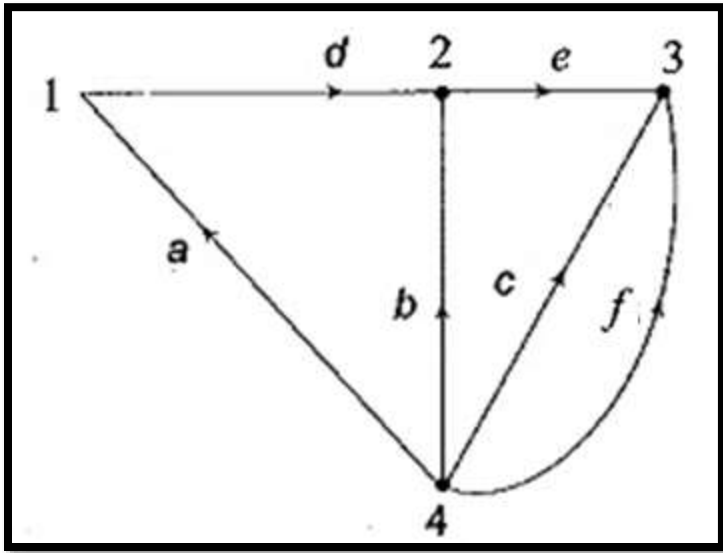


Fig: graph

From the graph: No. of nodes = 4 and
No. of branches = 6

The rows and columns of the above matrix represents the nodes and branches of given directed graph. Therefore the order of incidence matrix is 4×6 .

The **incidence matrix** corresponding to the above directed graph will be

$$A = \begin{array}{c} \text{Nodes} \\ \downarrow \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} \end{array} \begin{array}{c} \text{Branches} \rightarrow \\ \begin{array}{cccccc} a & b & c & d & e & f \end{array} \end{array} \left[\begin{array}{cccccc} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

Draw the oriented graph from the given incidence matrix?

Nodes	Branches					
	a	b	c	d	e	f
L	-1	0	0	1	-1	0
M	1	-1	0	0	0	-1
N	0	1	-1	0	1	0
O	0	0	1	-1	0	1

Nodes 4 $\rightarrow$ L, M, N and O

Branches 6 $\rightarrow$ a, b, c, d, e and f

Form Matrix, branch 'a' is connected between nodes L and M and orientation of branch 'a' at node L is -1 so it is coming towards node L .

while the orientation at node M is +1 so it is going away from node.

Similarly remaining can be drawn for the graph.

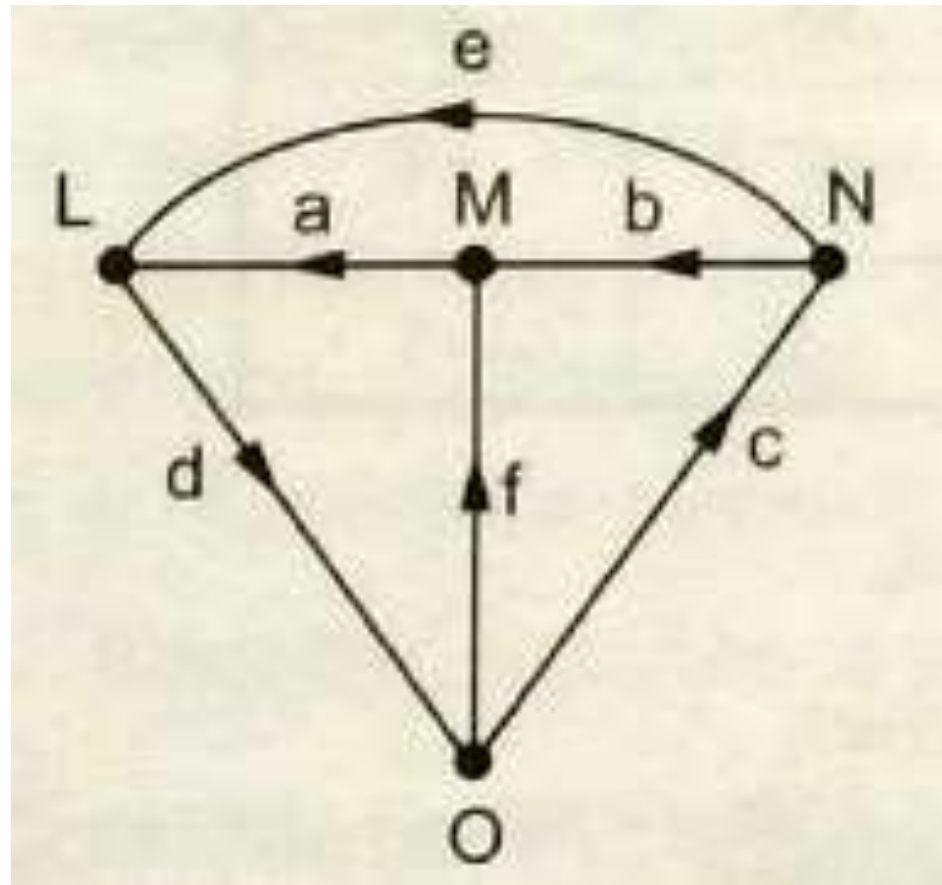


Fig: Oriented graph

So, the fundamental loop matrix will have 'b-n+1' rows and 'b' columns. Here, rows and columns are corresponding to the links of co-tree and branches of given graph.

Hence, the order of fundamental loop matrix will be $(b - n + 1) \times b$.

The **elements of fundamental loop matrix** will be having one of these three values, +1, -1 and 0.

The value of elements will be 0 for the remaining links and twigs, which are not part of the selected f-loop.

If the direction of twig current of selected f-loop is same as that of f-loop link current, then the value of element will be +1.

If the direction of twig current of selected f-loop is opposite to that of f-loop link current, then the value of element will be -1.

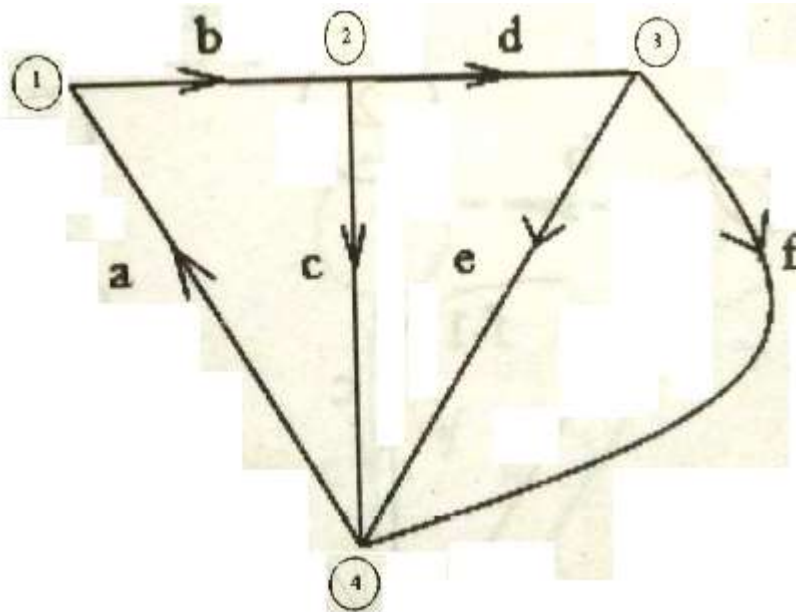
Procedure to find Fundamental Loop Matrix

Follow these steps in order to find the fundamental loop matrix of given directed graph.

1. Select a tree of given directed graph.
2. By including one link at a time, we will get one f-loop.
3. Fill the values of elements corresponding to this f-loop in a row of fundamental loop matrix.
4. Repeat the above step for all links.

RELATION BETWEEN LOOP CURRENTS AND BRANCH CURRENTS

Consider the graph shown below

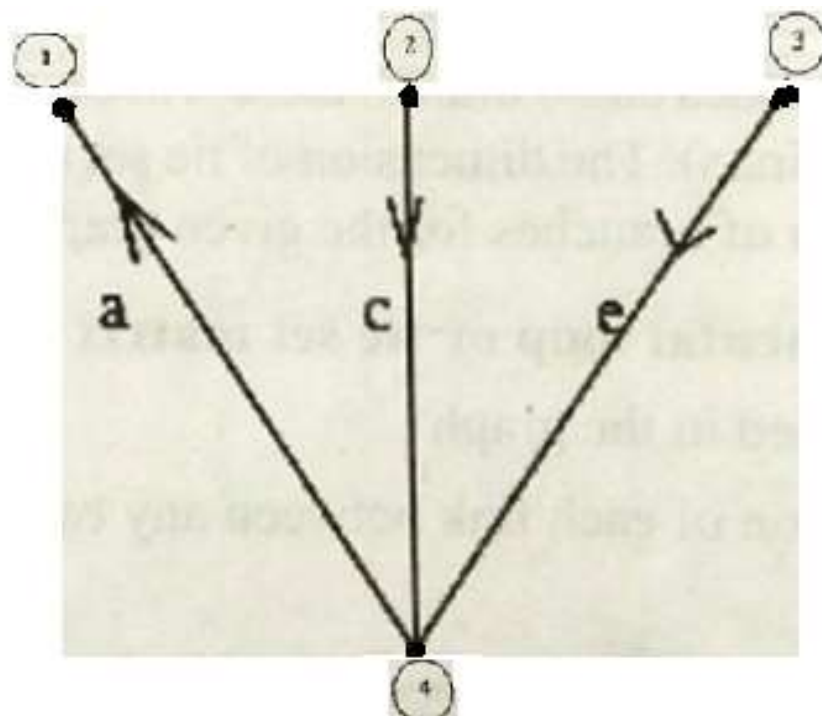


No of tree branches or twigs = $n - 1 = 4 - 1 = 3$

No of link branches $l = b - n + 1 = 6 - 4 + 1 = 3$

No of f-loops = no of tie sets = $b - n + 1 = 3$

Step 1: Arbitrarily select a tree graph shown in fig.



Set of twigs = {a, c, e}

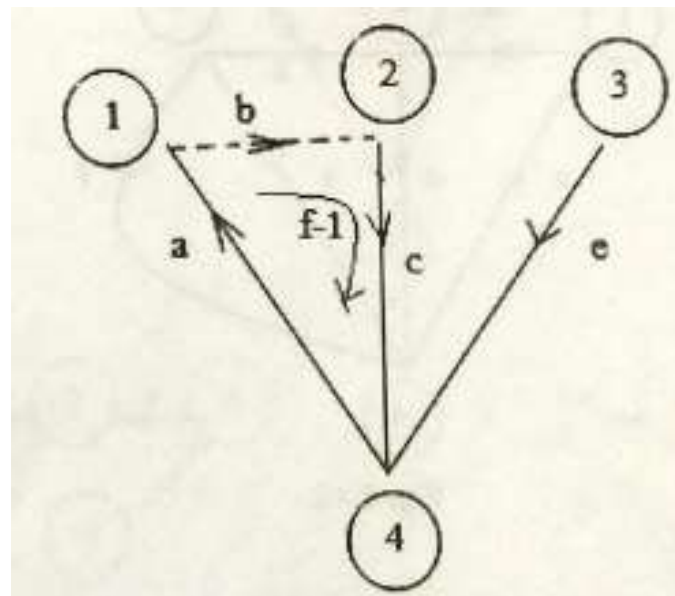
Set of links = {b, d, f}

Step 2: For a selected tree, addition of each link between any two nodes forms a loop called fundamental loop or tie set.

Assume branch voltage are represented by $V_a, V_b, V_c, V_d, V_e, V_f$

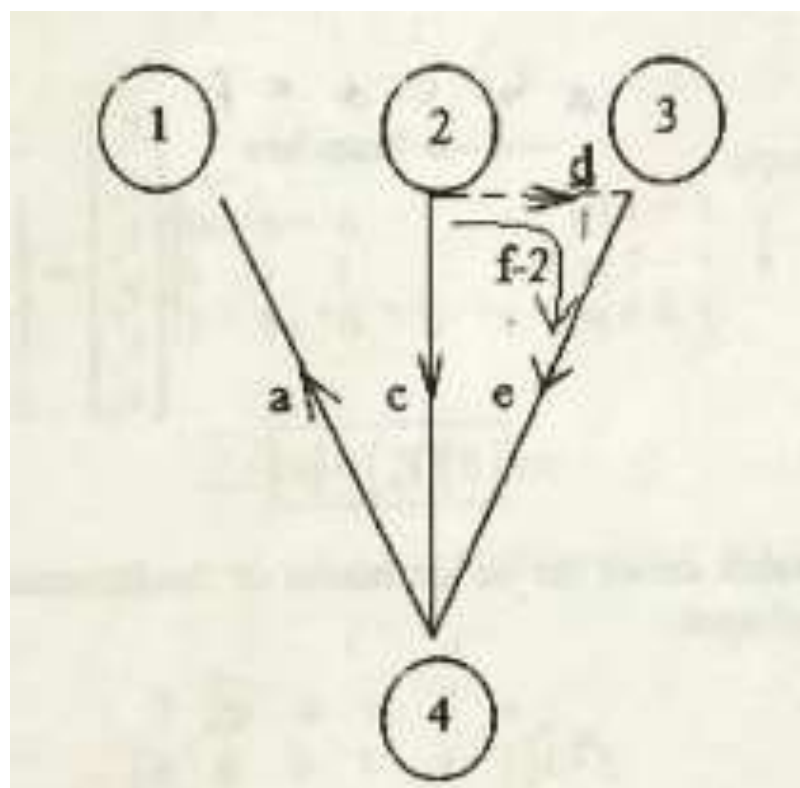
Addition of link 'b' in between nodes 1 and 2 form tie set -1 or fundamental loop-1 or f-loop-1,

with the orientation of fundamental loop-1 is coincide with orientation of link branch 'b'



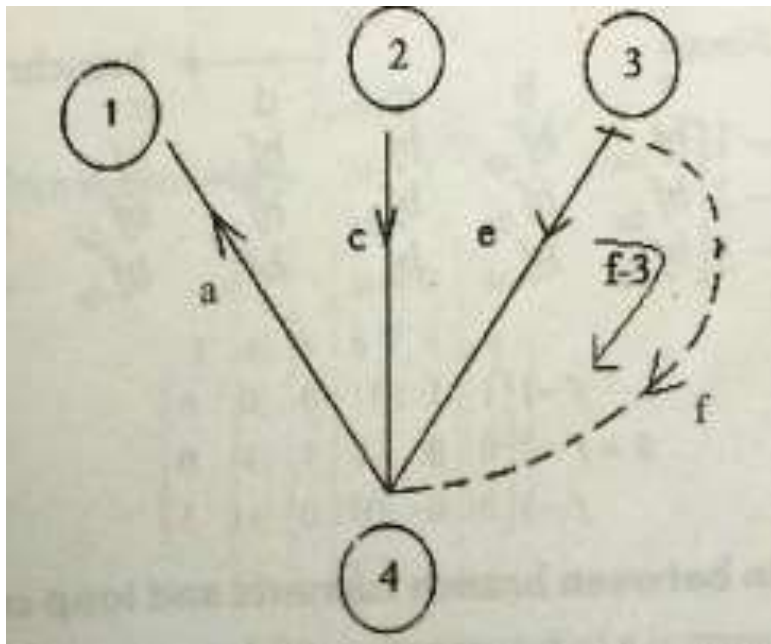
Tie Set-1 = {a,b,c}

Addition of link 'd' in between nodes 2 and 3 form tie set-2 or f-loop-2 and with the orientation of f-loop-2 is coincide with orientation of link branch 'd'



Tie Set-2 = {c,d,e}

Addition of link 'f' in between nodes 3 and 4 from tie set -3 or f loop-3 and with orientation of f-loop 3 is coincide with orientation of link branch 'f'



Tie Set-3 = {c, f}

Therefore the tie-set matrix is given from the above three tie-sets

f-Loops



$B =$

Branches $\rightarrow$

	a	b	c	d	e	f
$f-1$	1	1	1	0	0	0
$f-2$	0	0	-1	1	1	0
$f-3$	0	0	0	0	-1	1

It is possible to relate branch currents as a linear combination of link currents using matrix B. Since all the branches are connected to all the nodes in the graph.

Let the branch currents be i_a , i_b , i_c , i_d , i_e & i_f .

There are three links and let the link currents be I_a , I_b , I_c for links b, d and f respectively.

Each branch currents can be expressed in terms of link branch current as follows

$$i_b = I_b$$

$$i_d = I_d$$

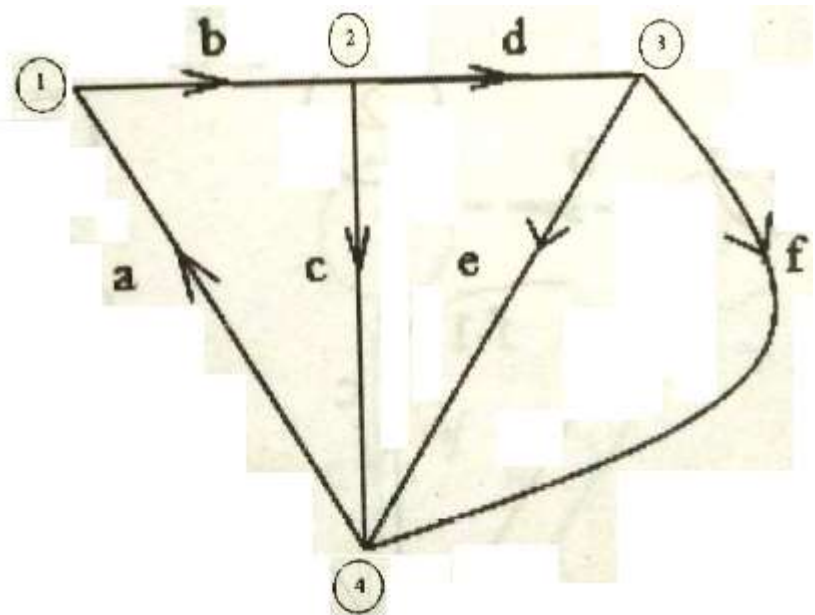
$$i_f = I_f$$

Applying KCL at node 1

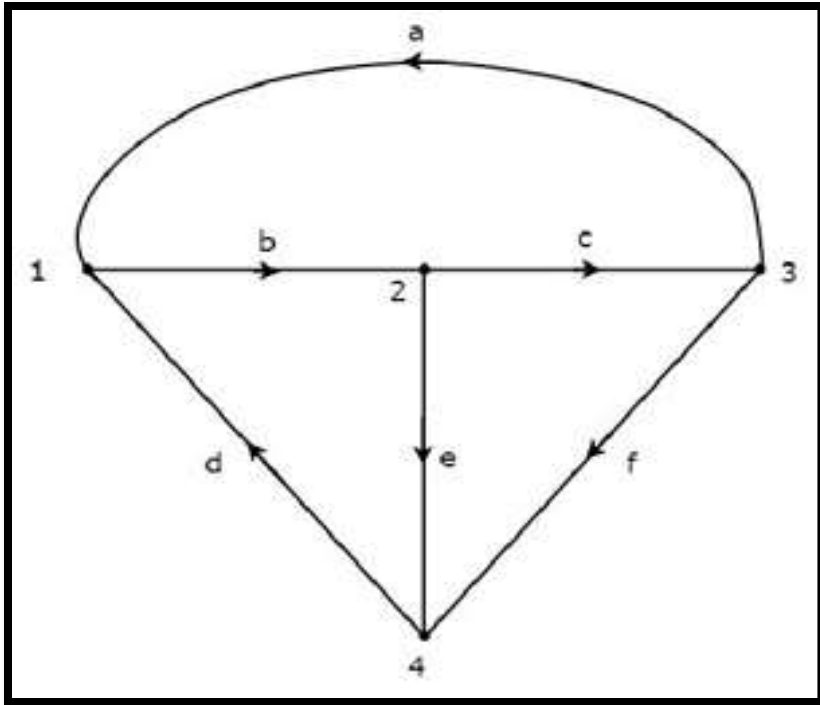
$$\begin{aligned} -i_a + i_b &= 0 \\ \Rightarrow i_a &= i_b = I_b \end{aligned}$$

Applying KCL at node 2

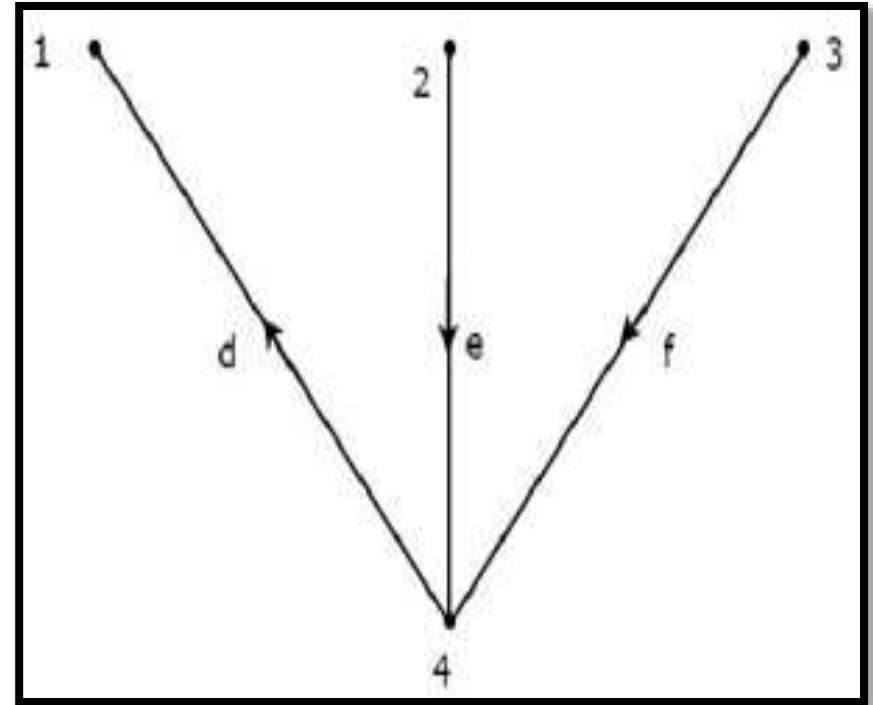
$$\begin{aligned} -i_b + i_c + i_d &= 0 \\ \Rightarrow i_c &= i_b - i_d = I_b - I_d \end{aligned}$$



Obtain the incidence matrix for the directed graph and its selected tree shown in figure.



(a) Directed Graph



(b) Selected Tree

The given Tree contains three branches d, e & f.

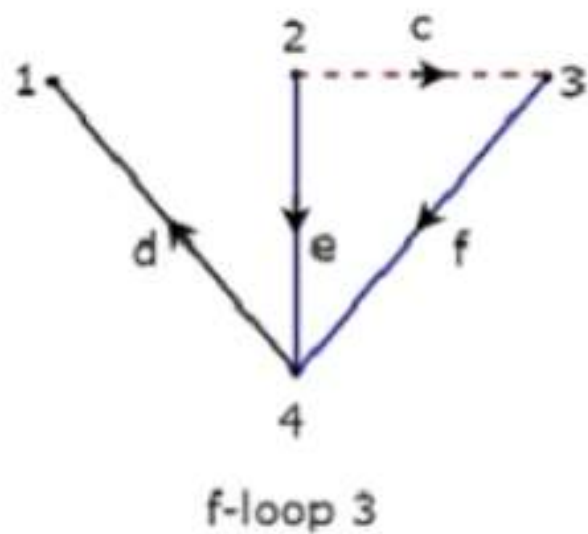
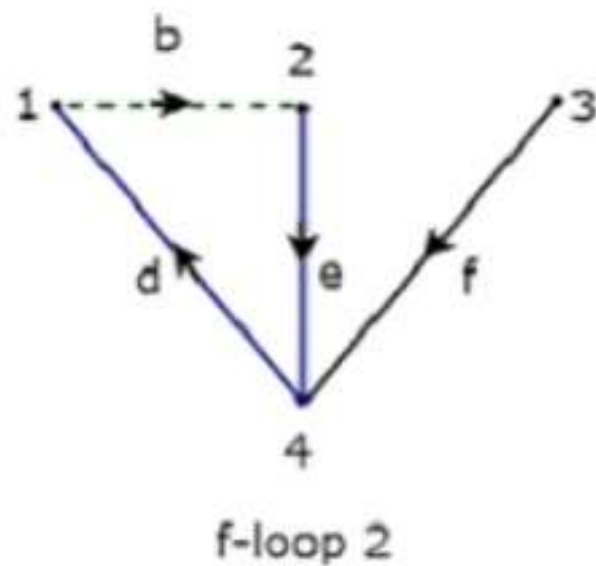
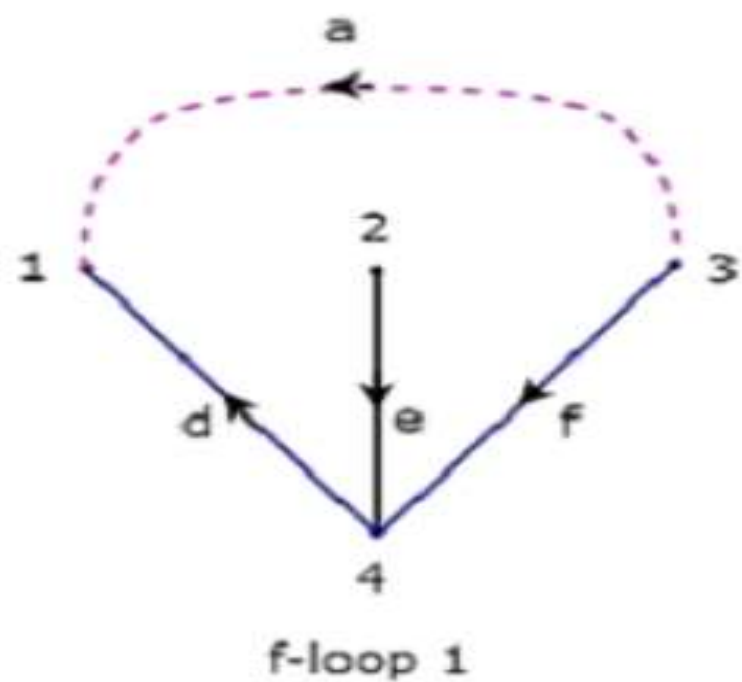
Hence, the branches a, b & c will be the links of the Co-Tree corresponding to the given Tree.

By including one link at a time to the above Tree, we will get one **f-loop**.

So, there will be three **f-loops**, since there are three links.

These three f-loops are shown in the following figure.

Branches, which are represented with dotted lines (links) will forms the fundamental-loops.



We will get the row wise element values of Tie-set matrix from each f-loop.

So, the **Tie-set matrix** of the above considered Tree will be

$$B = \begin{array}{c} \downarrow \\ \text{f-Loops} \end{array} \begin{array}{c} \text{Branches} \rightarrow \\ a \quad b \quad c \quad d \quad e \quad f \end{array} \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

The rows and columns of the above matrix represents the links and branches of given directed graph.

The order of this incidence matrix is 3×6 .

Note: The **number of Fundamental loop matrices** of a directed graph will be equal to the number of Trees of that directed graph. Because, every Tree will be having one Fundamental loop matrix.

Fundamental Cut-set Matrix

Fundamental cut set or **f-cut set** is the minimum number of branches that are removed from a graph in such a way that the original graph will become two isolated subgraphs.

The f-cut set contains only **one twig** and one or more links.

So, the number of f-cut sets will be equal to the number of twigs.

Fundamental cut set matrix is represented with letter C.

This matrix gives the relation between branch voltages and twig voltages.

If there are 'n' nodes and 'b' branches are present in a **directed graph**, then the number of twigs present in a selected Tree of given graph will be $n-1$.

So, the fundamental cut set matrix will have ' $n-1$ ' rows and 'b' columns.

Here, rows and columns are corresponding to the twigs of selected tree and branches of given graph. Hence, the **order** of fundamental cut set matrix will be **$(n-1) \times b$** .

The **elements of fundamental cut set matrix** will be having one of these three values, +1, -1 and 0.

The value of elements will be 0 for the remaining twigs and links, which are not part of the selected f-cut-set.

If the direction of link current of selected f-cut set is same as that of f-cut-set twig current, then the value of element will be +1.

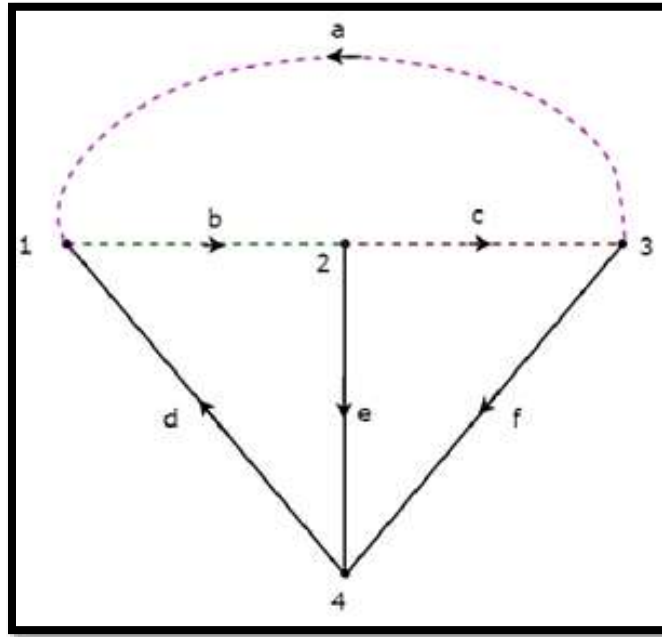
If the direction of link current of selected f-cut set is opposite to that of f-cut-set twig current, then the value of element will be -1.

Procedure to find Fundamental Cut-set Matrix

Follow these steps in order to find the fundamental cut set matrix of given directed graph.

1. Select a Tree of given directed graph and represent the links with the dotted lines.
2. By removing one twig and necessary links at a time, we will get one f-cut set.
3. Fill the values of elements corresponding to this f-cut set in a row of fundamental cut set matrix.
4. Repeat the above step for all twigs.

Obtain the cut-set matrix for the following tree of a graph

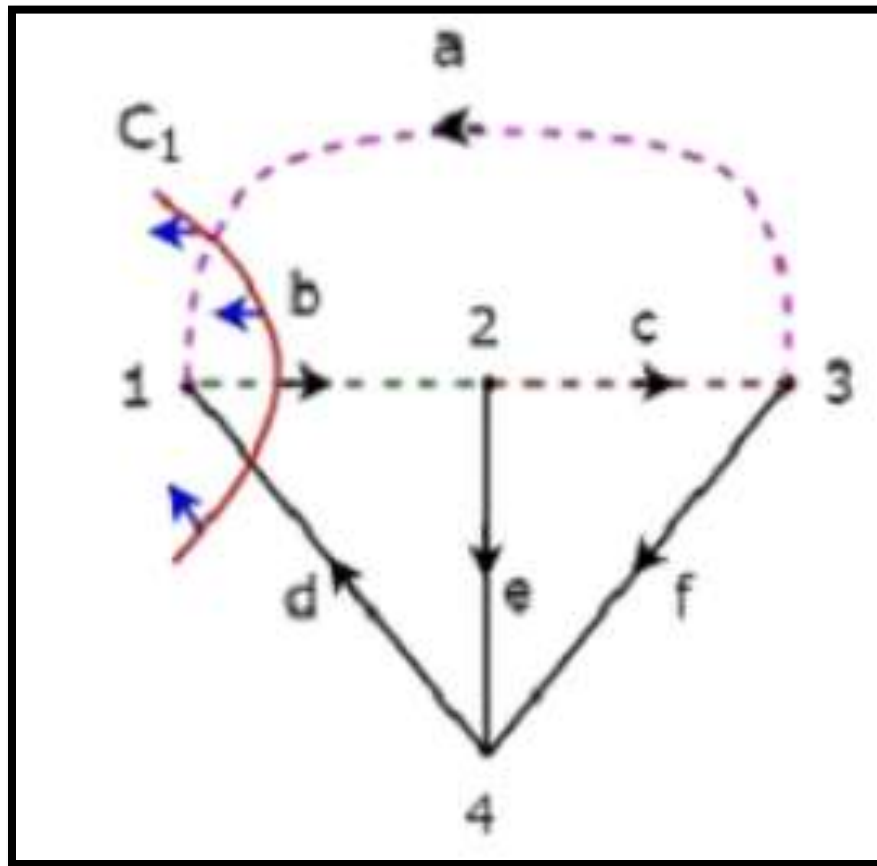


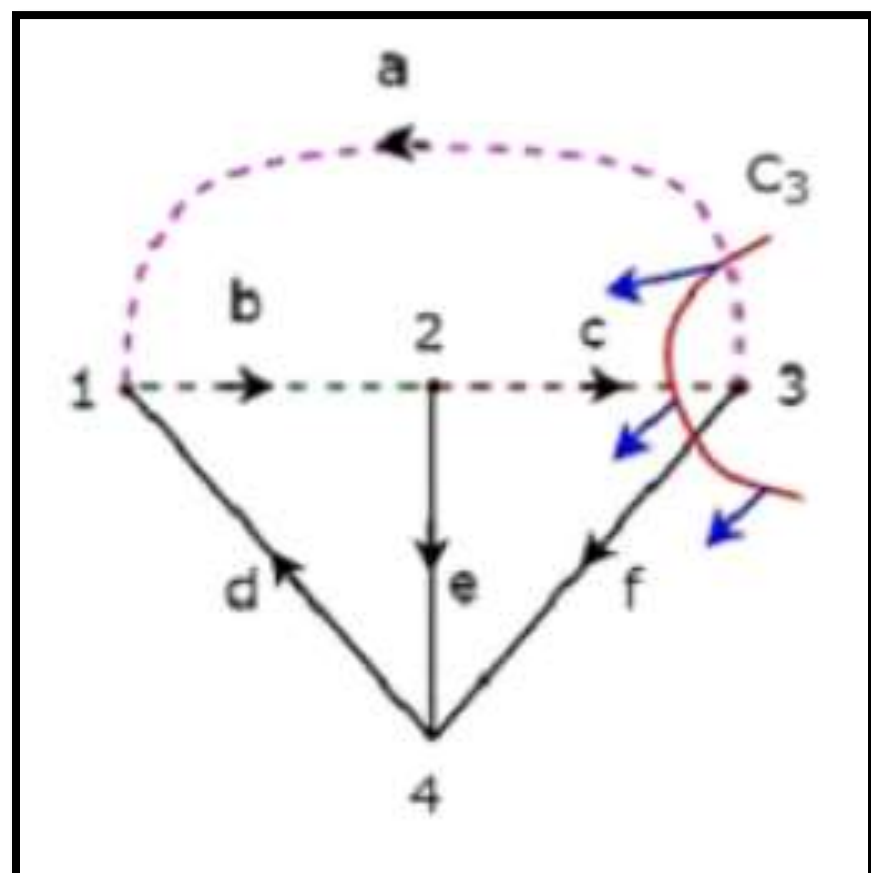
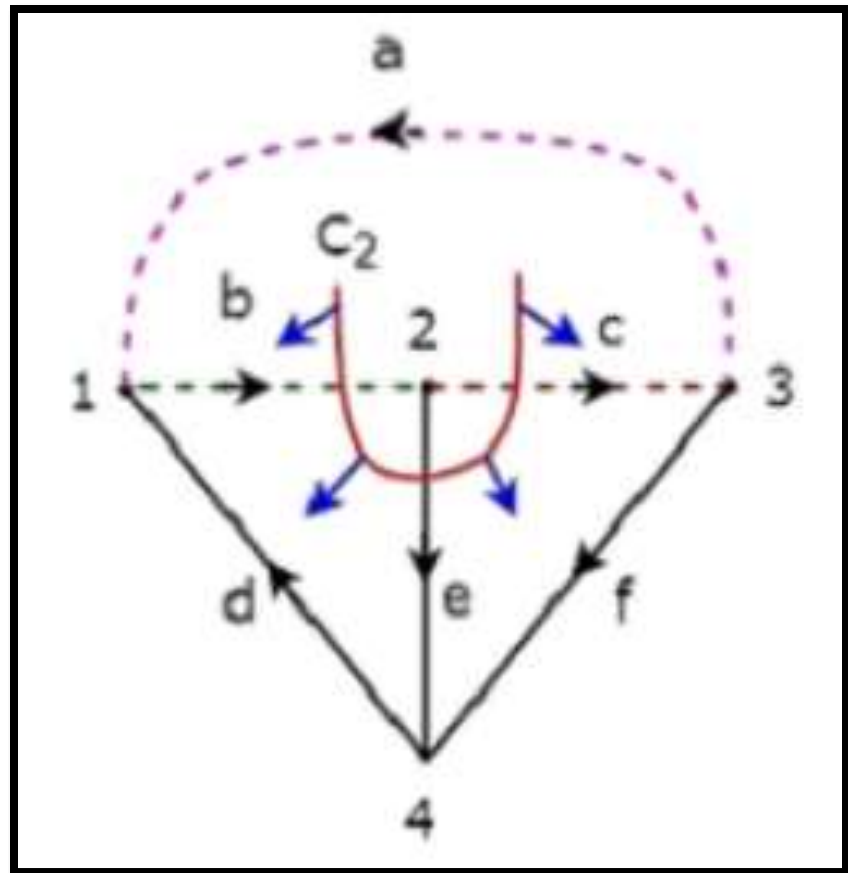
Selected the branches d, e & f of the directed graph as twigs. So, the remaining branches a, b & c of this directed graph will be the links.

The **twigs** d, e & f are represented with solid lines and **links** a, b & c are represented with dotted lines in the following figure.

By removing one twig and necessary links at a time, we will get one f-cut set.

So, there will be three f-cut sets, since there are three twigs. These three **f-cut sets** are shown in the following figure.





We will be having three f-cut sets by removing a set of twig and links of C_1 , C_2 and C_3 . We will get the row wise element values of fundamental cut set matrix from each f-cut set.

So, the **fundamental cut set matrix** of the above considered Tree will be

$$\begin{array}{c}
 \text{Twigs} \\
 \downarrow \\
 C =
 \end{array}
 \begin{array}{c}
 \text{Branches} \rightarrow \\
 \begin{matrix} a & b & c & d & e & f \end{matrix} \\
 \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 & 0 & 1 \end{bmatrix}
 \end{array}$$

The rows and columns of the above matrix represents the twigs and branches of given directed graph. The order of this fundamental cut set matrix is 3×6 .

Note: The **number of Fundamental cut set matrices** of a directed graph will be equal to the number of Trees of that directed graph. Because, every Tree will be having one Fundamental cut set matrix.

In the following network, the numerical values of resistances also indicate the branch numbers. Write the oriented graph of the network. Select a tree with branches 1, 2, 3 as the tree branches, write tie-set and cut-set matrices.

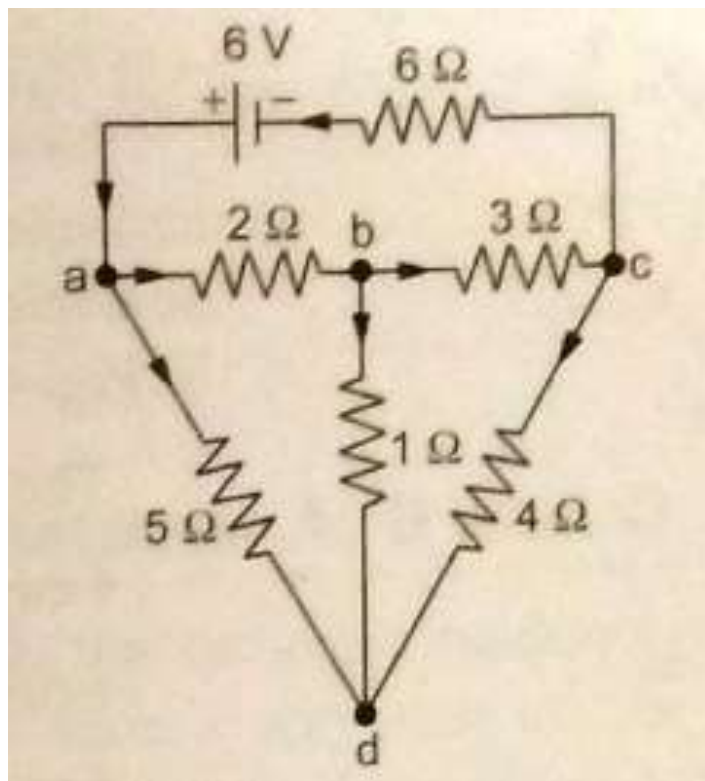


Fig: Given Network

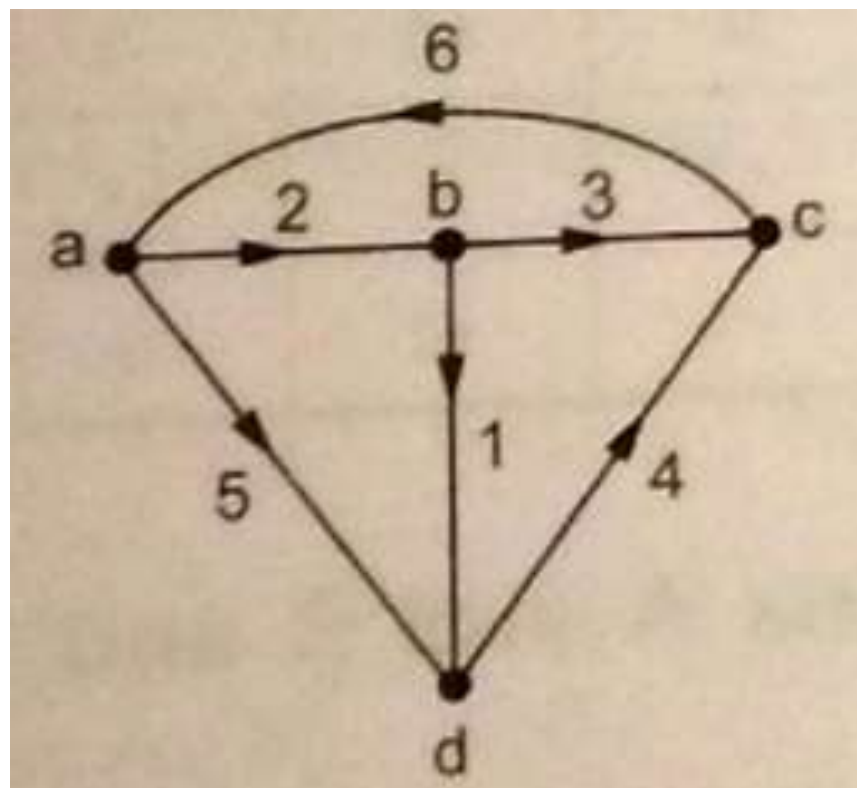


Fig: Oriented graph

The tiesets are written as follows,

f-tieset 1 : I_A [5, 1, 2]

f-tieset 2 : I_B [4, 3, 1]

f-tieset 3 : I_C [6, 2, 3]

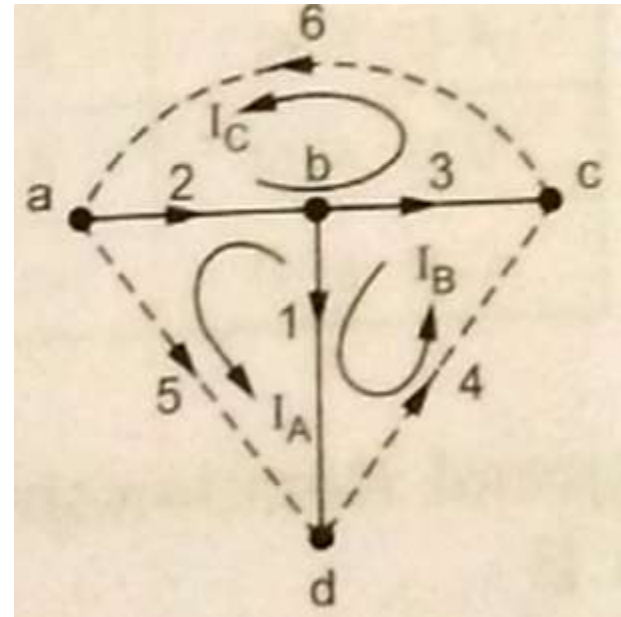


Fig: Tree of a graph

The tieset schedule can be written as follows,

Tiesets	Branches					
	1	2	3	4	5	6
I_A [5, 1, 2]	- 1	- 1	0	0	+ 1	0
I_B [4, 3, 1]	+ 1	0	- 1	+ 1	0	0
I_C [6, 2, 3]	0	+ 1	+ 1	0	0	+ 1

The cutsets are written as follows,

f-cutset 1 : A [2, 5, 6]

f-cutset 2 : B [3, 4, 6]

f-cutset 3 : C [1, 5, 4]

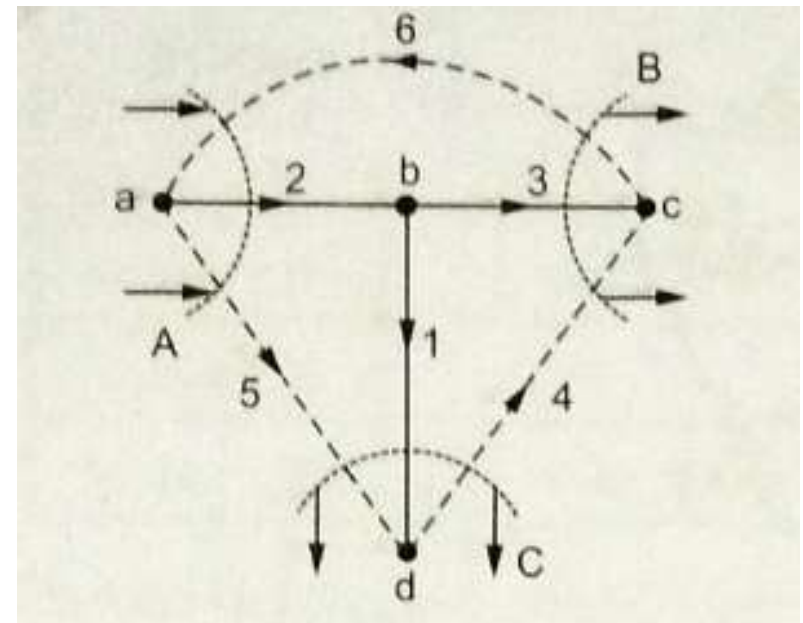


Fig: Tree of a graph

The cutset schedule is written as follows :

f-cutsets	Branches					
	1	2	3	4	5	6
A [2, 5, 6]	0	+1	0	0	+1	-1
B [3, 4, 6]	0	0	+1	+1	0	-1
C [1, 5, 4]	+1	0	0	-1	+1	0

Dual networks:

Two networks are said to be dual networks of each other if the mesh equations of given network are the node equations of other network.

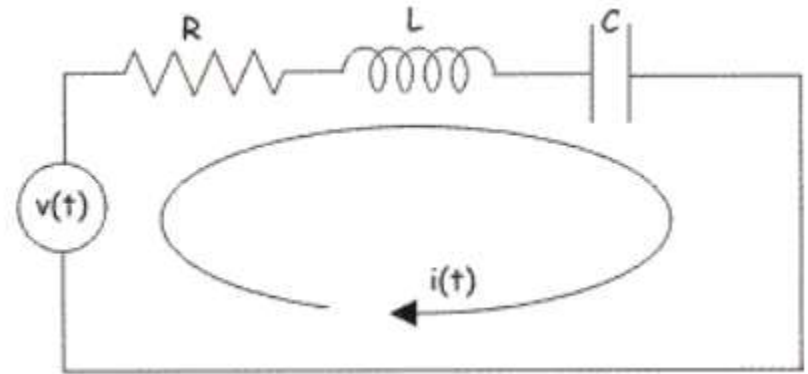
Dual networks are based on Kirchhoff's voltage and current laws.

Consider simple series RLC circuit excited by AC voltage as shown in figure.

Applying KVL equation,

$$v(t) = Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt$$

.....(1)



Consider parallel RLC circuit

Applying KCL to the circuit

$$i(t) = i_1(t) + i_2(t) + i_3(t)$$

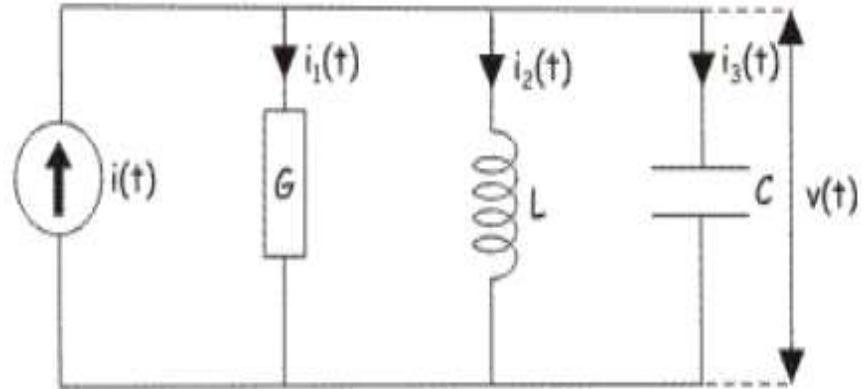
$$i(t) = Gv(t) + C \frac{dv(t)}{dt} + \frac{1}{L} \int v(t) dt$$

.....(2)

Thus equations (1) and (2) are similar.

The voltage source V of the series RLC circuit corresponds to current source I in the parallel RLC circuit.

Also the current in the first circuit corresponds to voltage in second network.



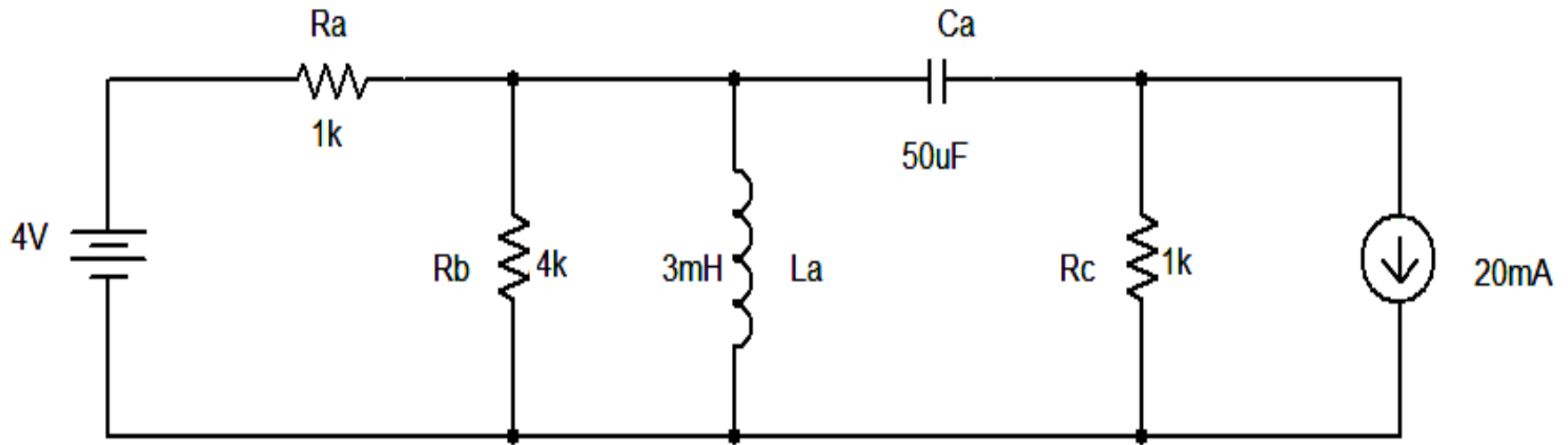
List of element and its dual

Sr. No.	Element	Dual Element
1	Resistance	Conductance
2	Capacitance	Inductance
3	Inductance	Capacitance
4	Series Branch	Parallel Branch
5	Mesh	Node
6	Closed Switch	Open Switch

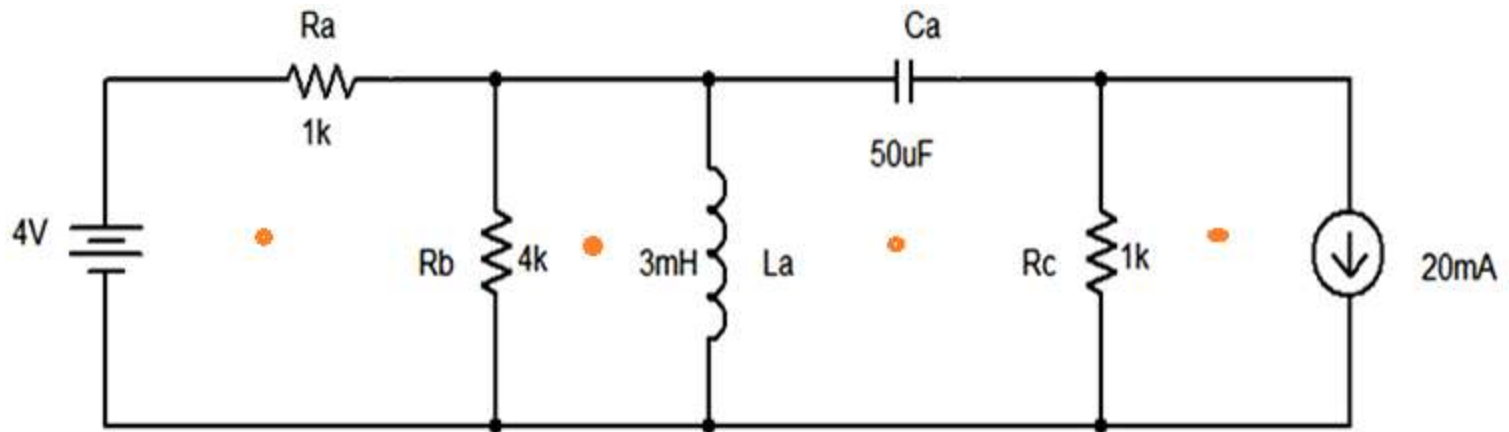
Rules to Construct Dual Circuits

1. Place a node at the center of each mesh of the circuit.
2. Place a reference node (ground) outside of the circuit.
3. Draw lines between nodes such that each line crosses an element.
4. Replace the element by its dual pair.
5. Determine the polarity of the voltage source and direction of the current source.

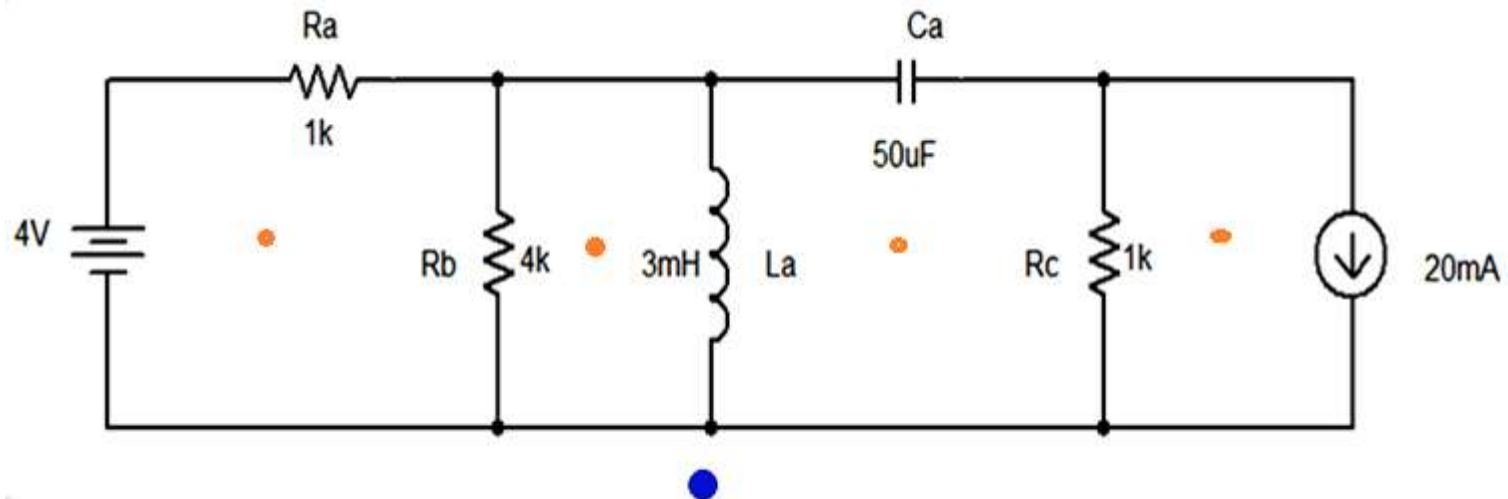
Draw the dual of the following network shown in the fig.



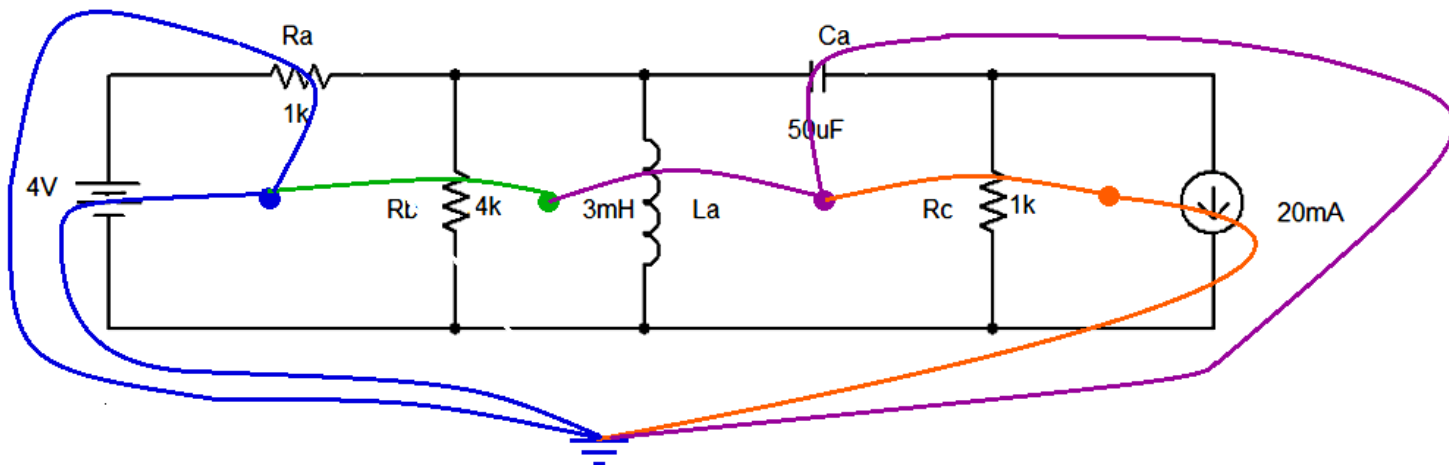
Step-1: Place a node at the center of each mesh of the circuit.



Step 2: Place a reference node (ground) outside of the circuit.



Step 3: Draw lines between nodes such that each line crosses an element.



Step 4: Replace the element by its dual pair.

Step 5: Determine the polarity of the voltage source and direction of the current source.

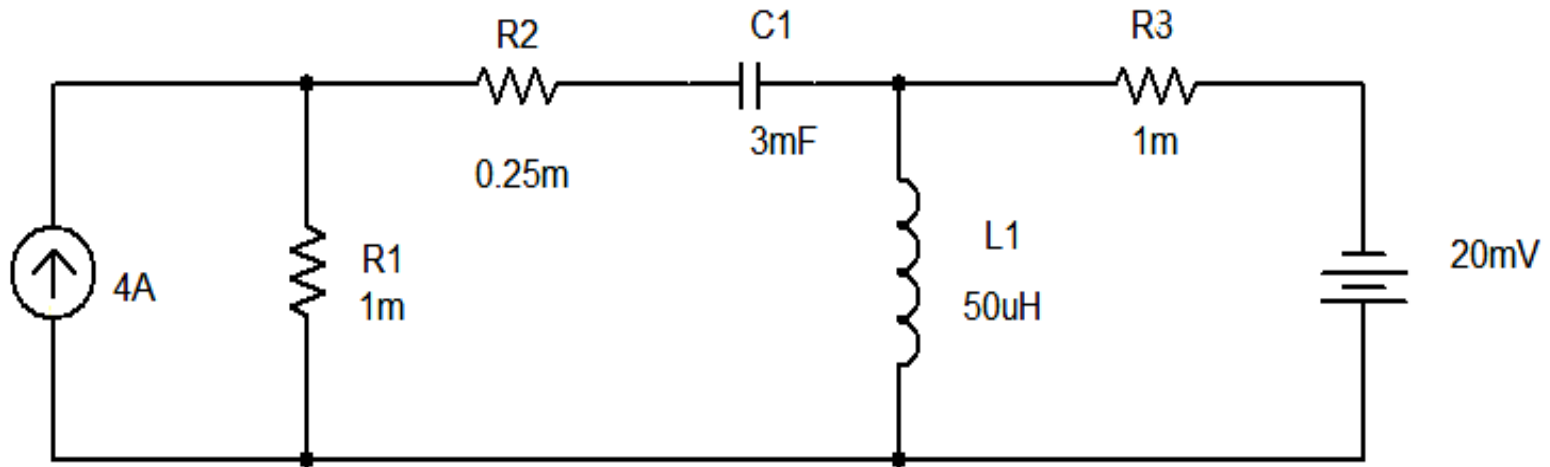


Fig: Draw Dual circuit

Draw the dual network for the given network shown in figure.

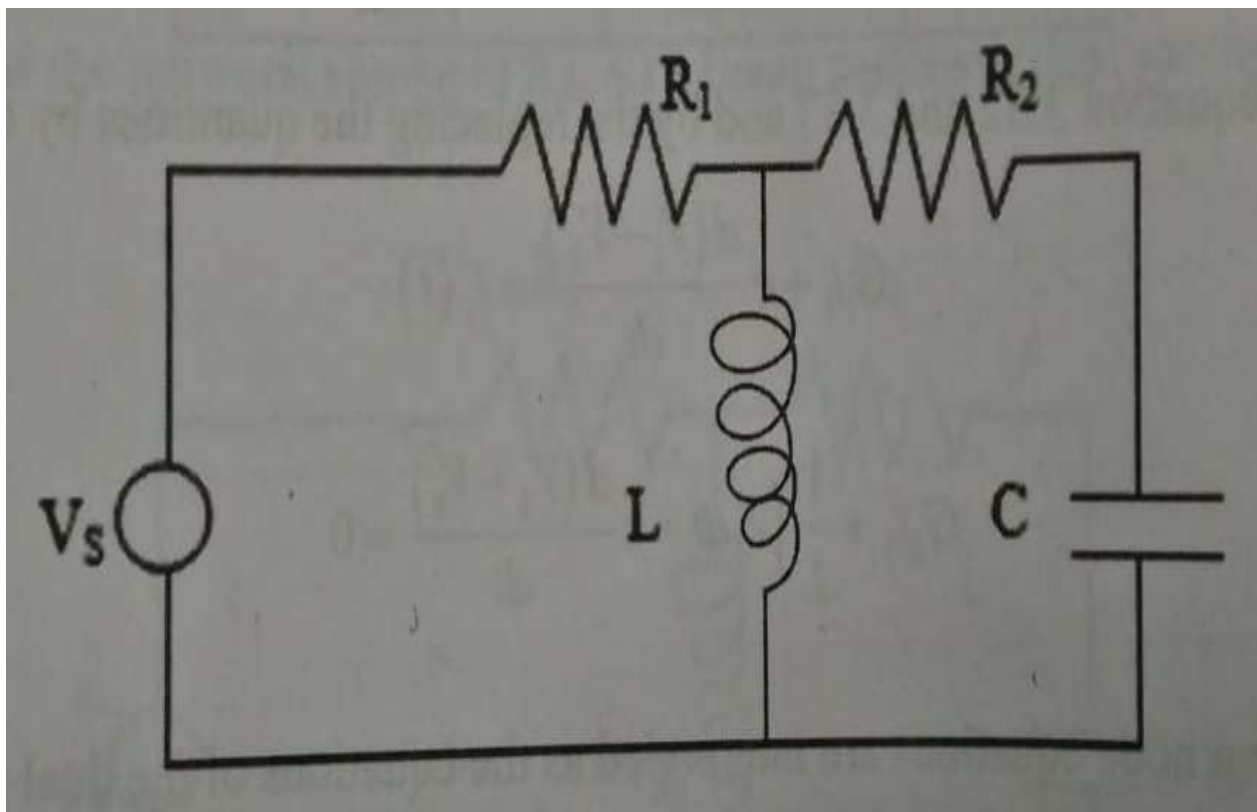
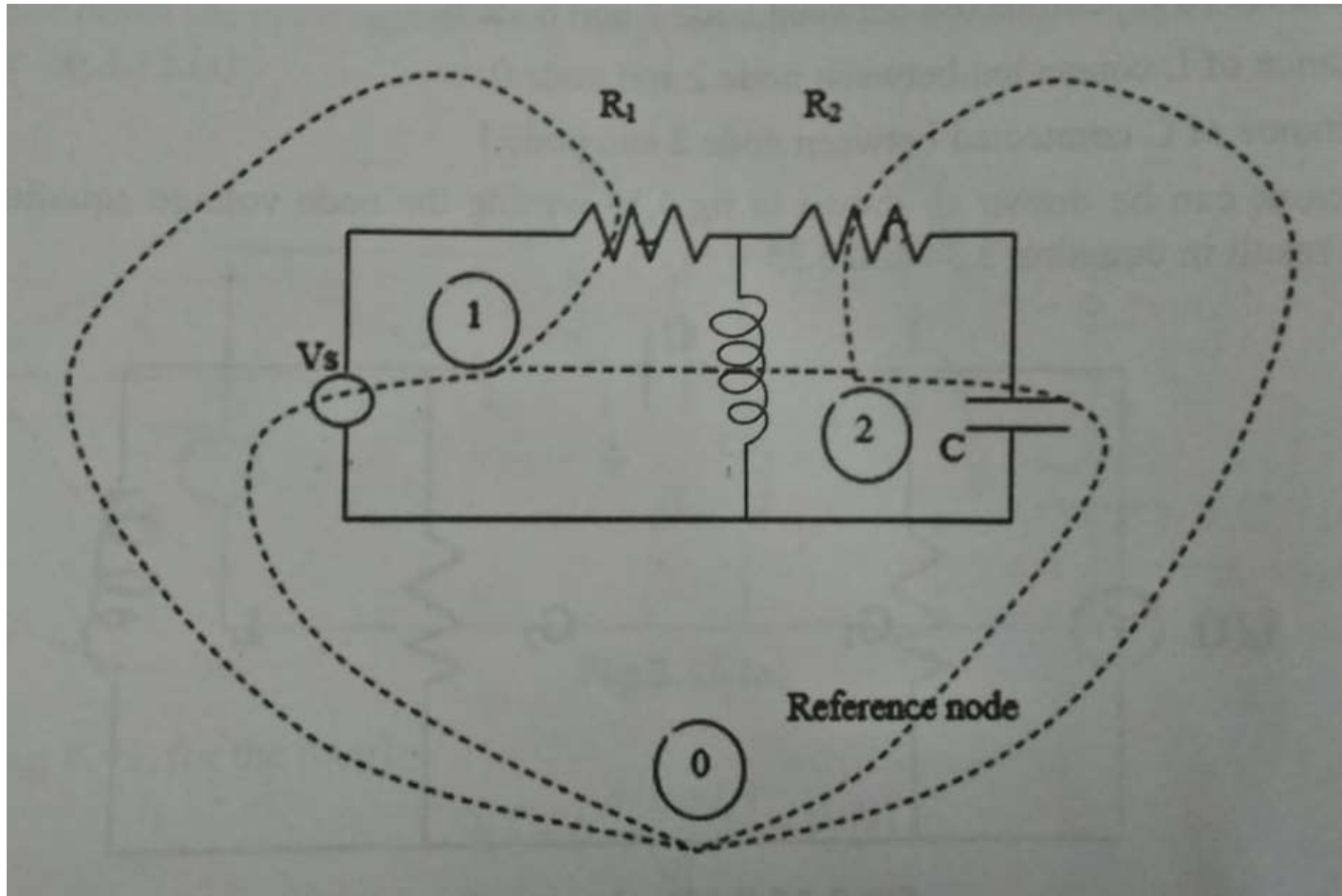
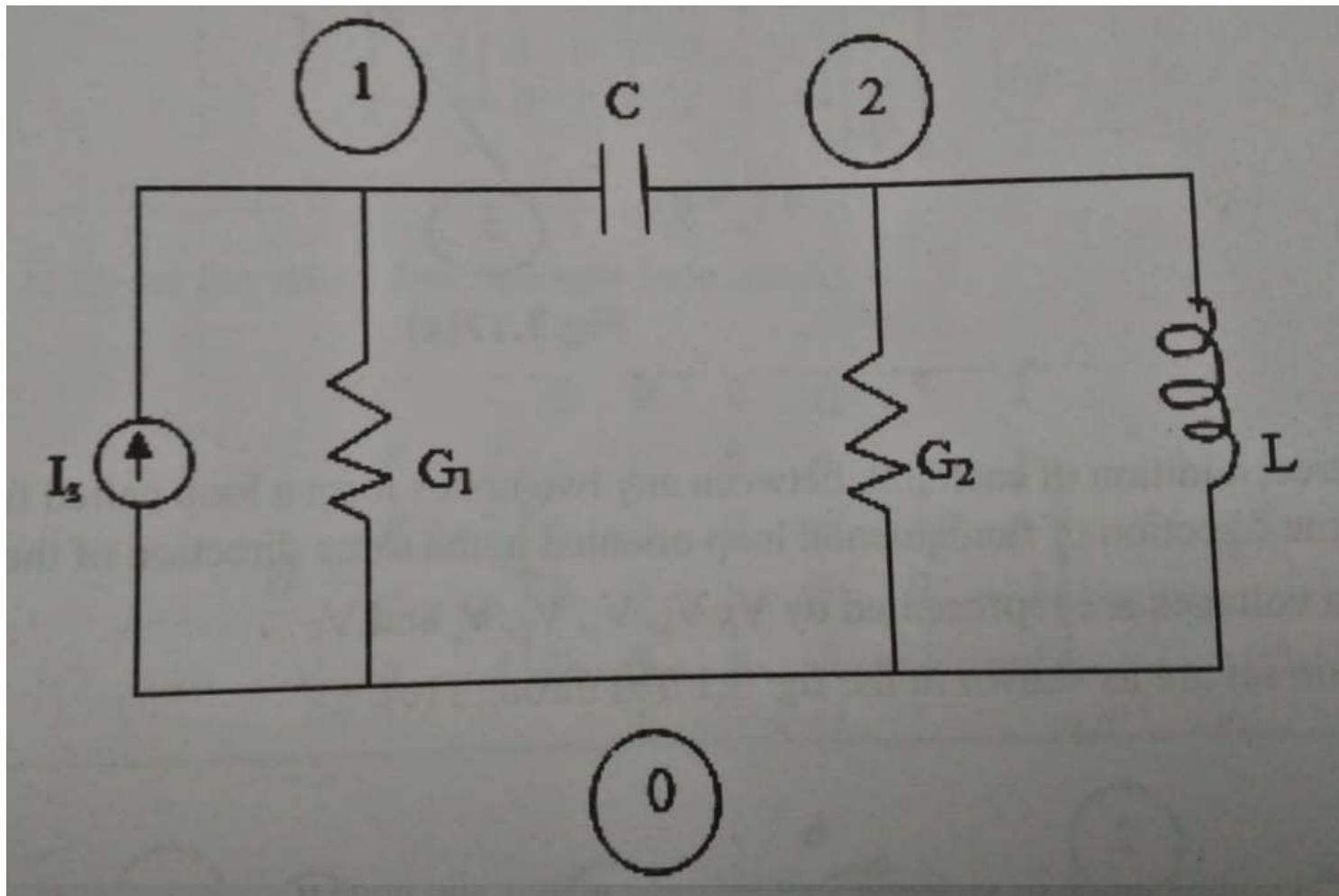


Fig: Network

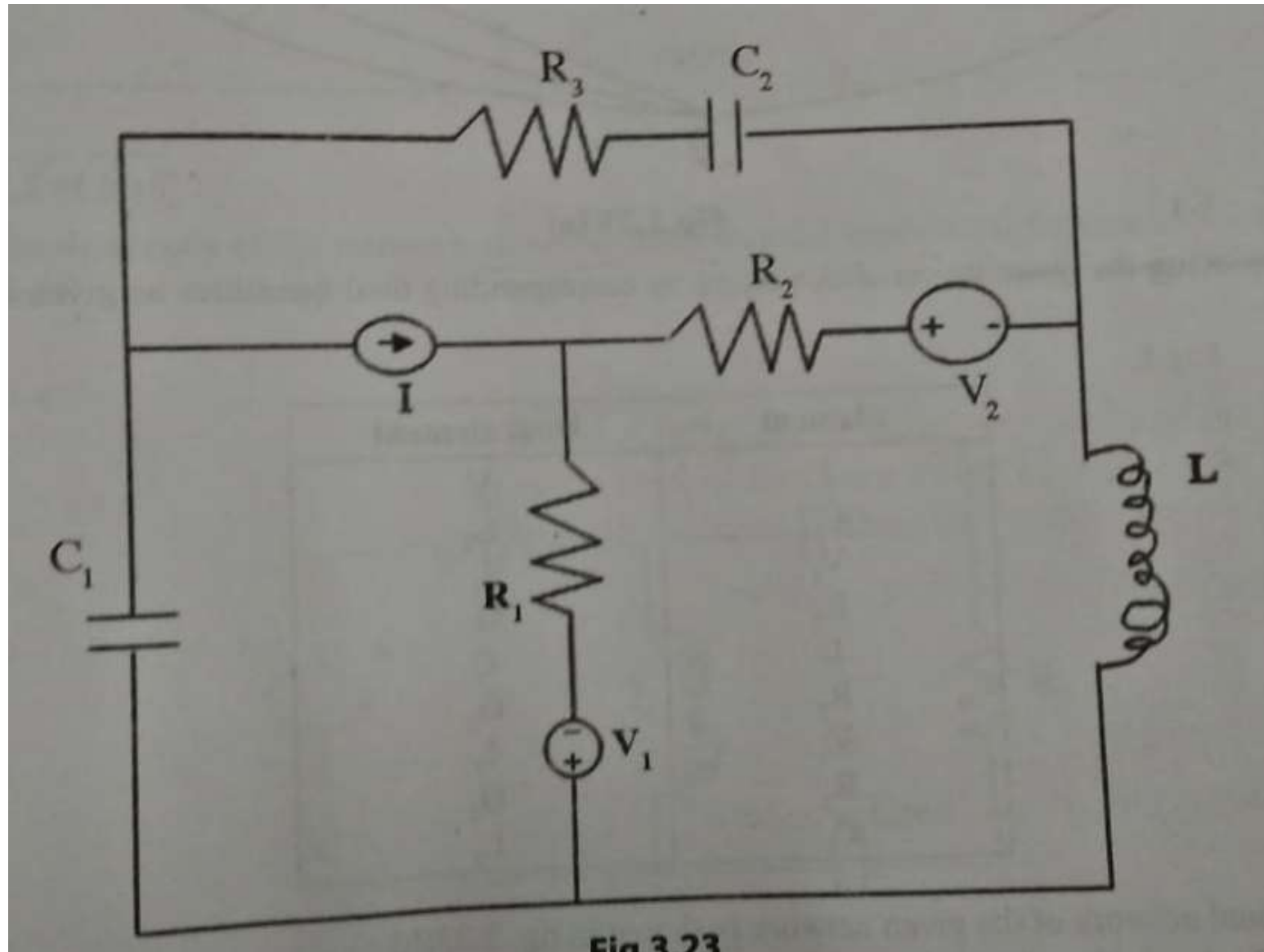
Place Dots in each mesh and also place a reference (datum) node outside of the network



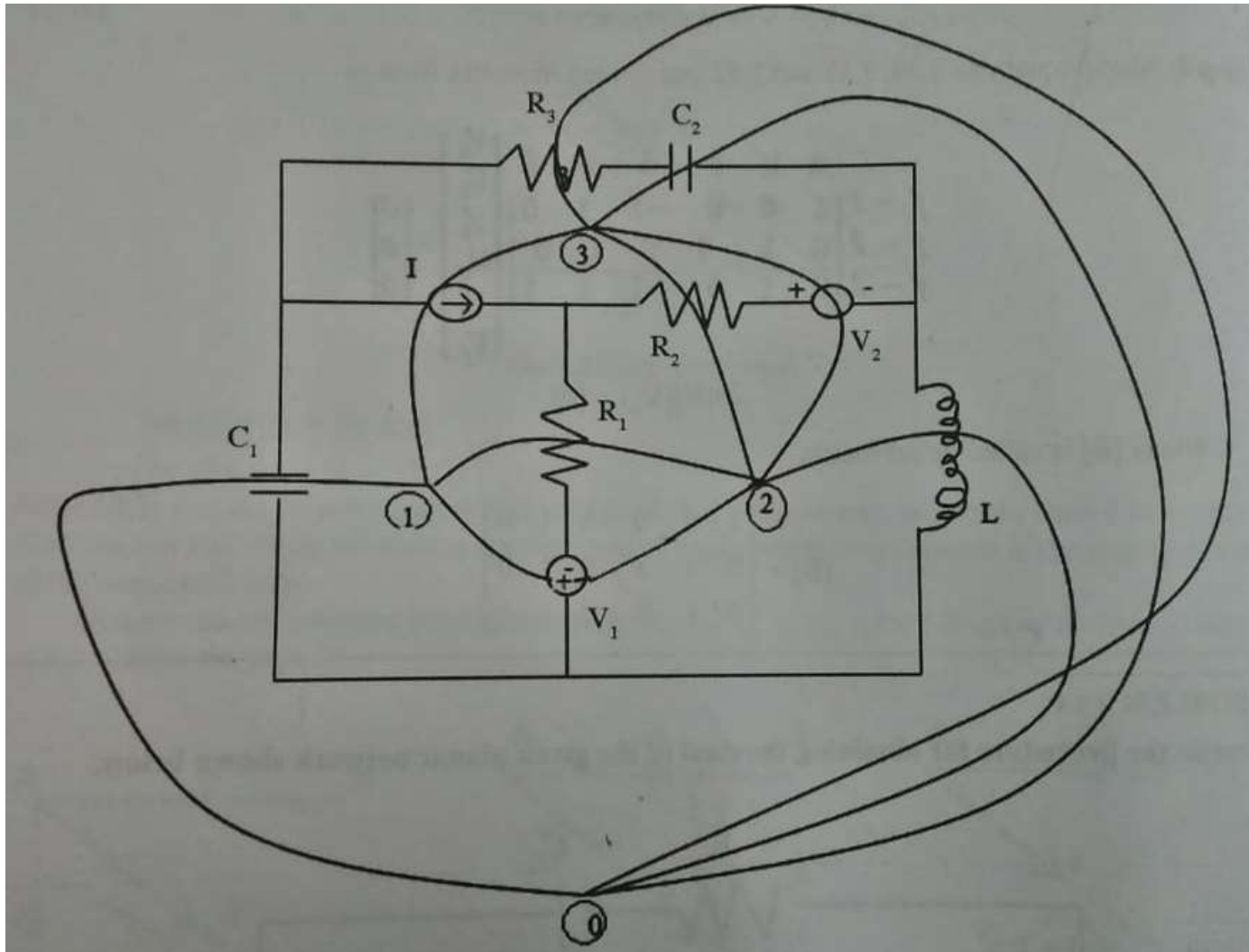
Draw the dual network by replacing the elements of the network with their dual elements



Draw the dual network for the given network shown in figure.



Place Dots in each mesh and also place a reference (datum) node outside of the network



Draw the dual network by replacing the elements of the network with their dual elements

